

Artificial Intelligence Methods

Computational Problem Solving

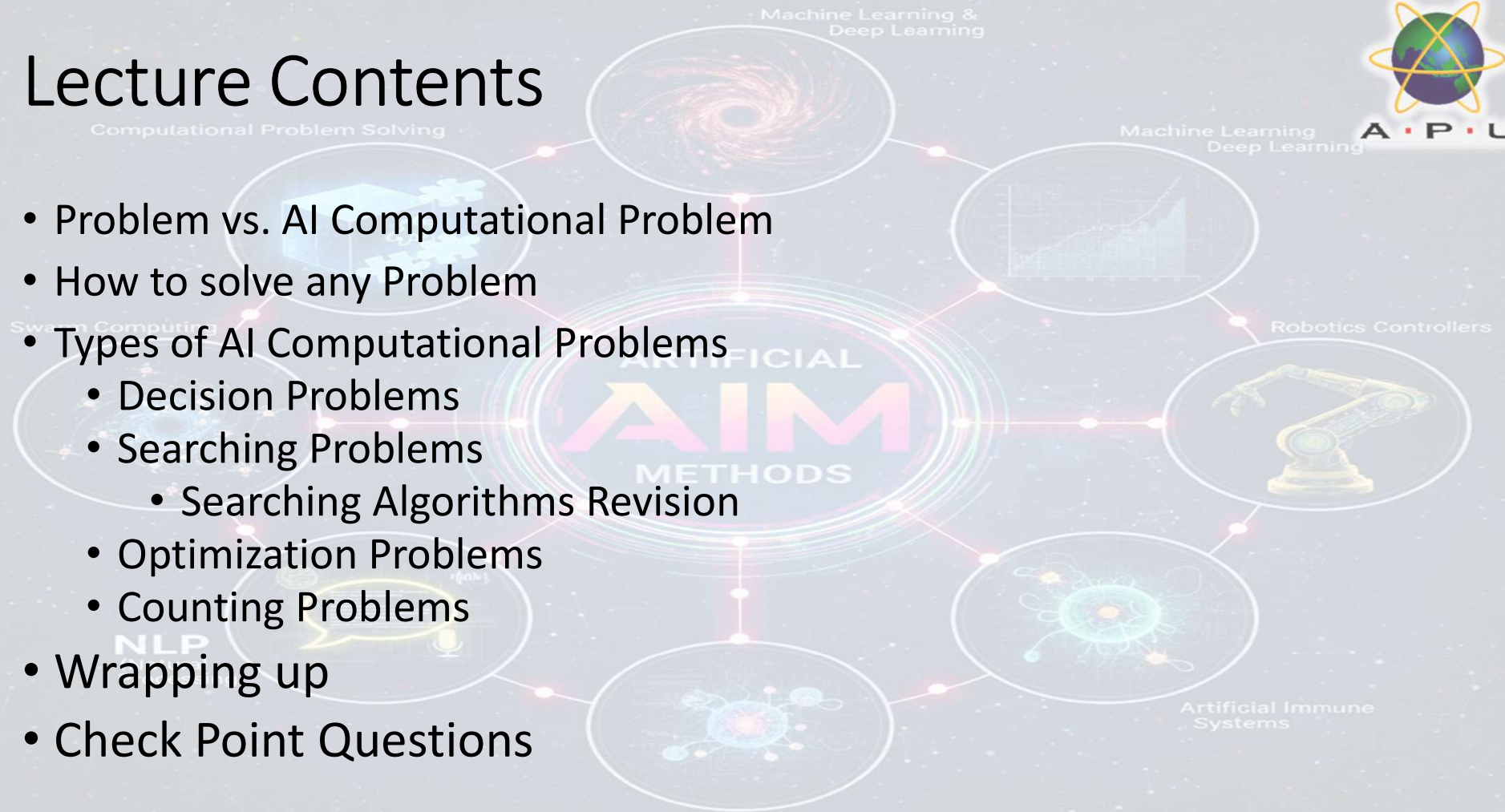
ARTIFICIAL
METHODS

By

Asst. Prof. Ts. Dr. Maythem Kamal Abbas Al-Adilee

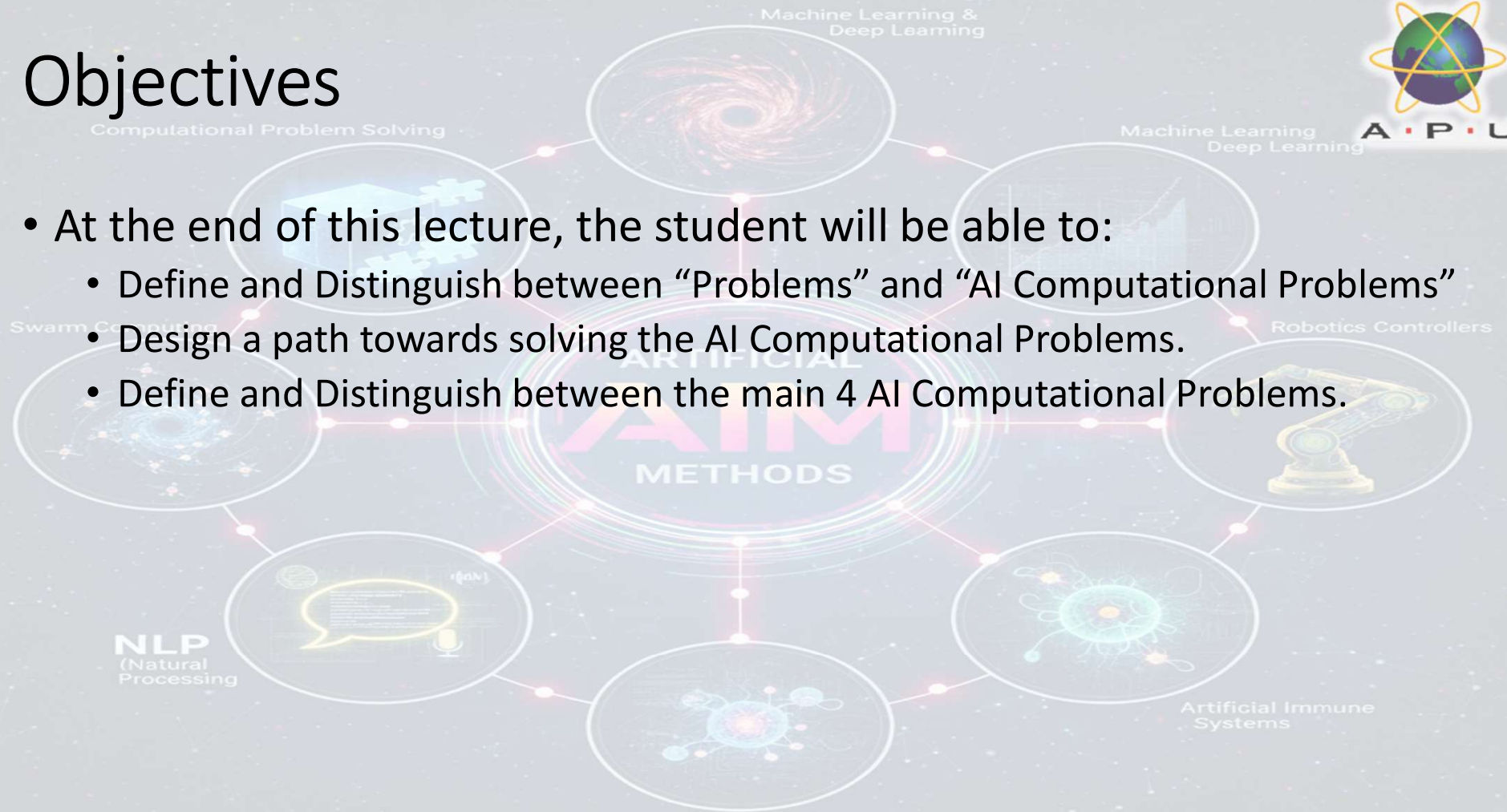
Lecture Contents

- Problem vs. AI Computational Problem
- How to solve any Problem
- Types of AI Computational Problems
 - Decision Problems
 - Searching Problems
 - Searching Algorithms Revision
 - Optimization Problems
 - Counting Problems
- Wrapping up
- Check Point Questions



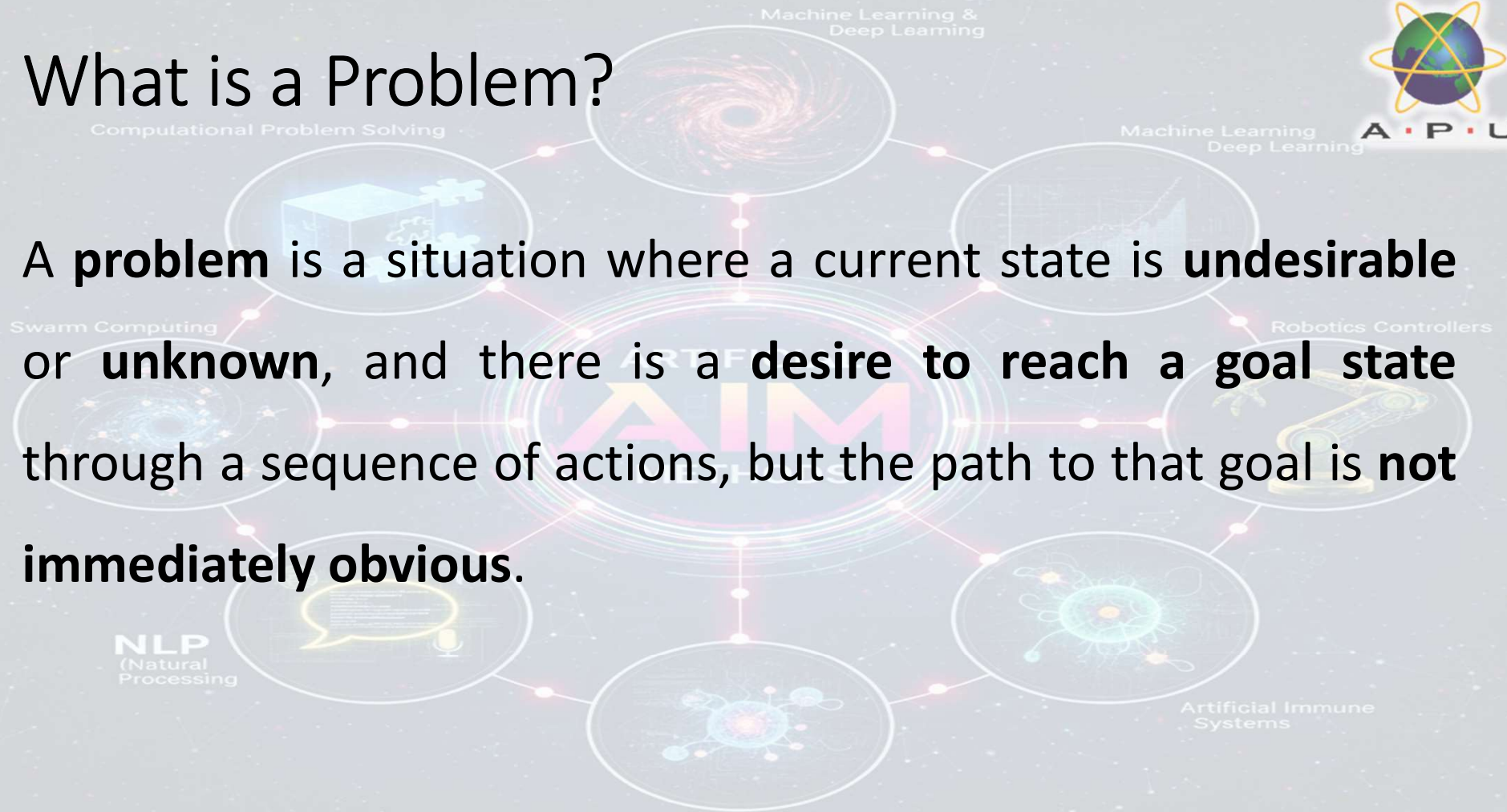
Objectives

- At the end of this lecture, the student will be able to:
 - Define and Distinguish between “Problems” and “AI Computational Problems”
 - Design a path towards solving the AI Computational Problems.
 - Define and Distinguish between the main 4 AI Computational Problems.

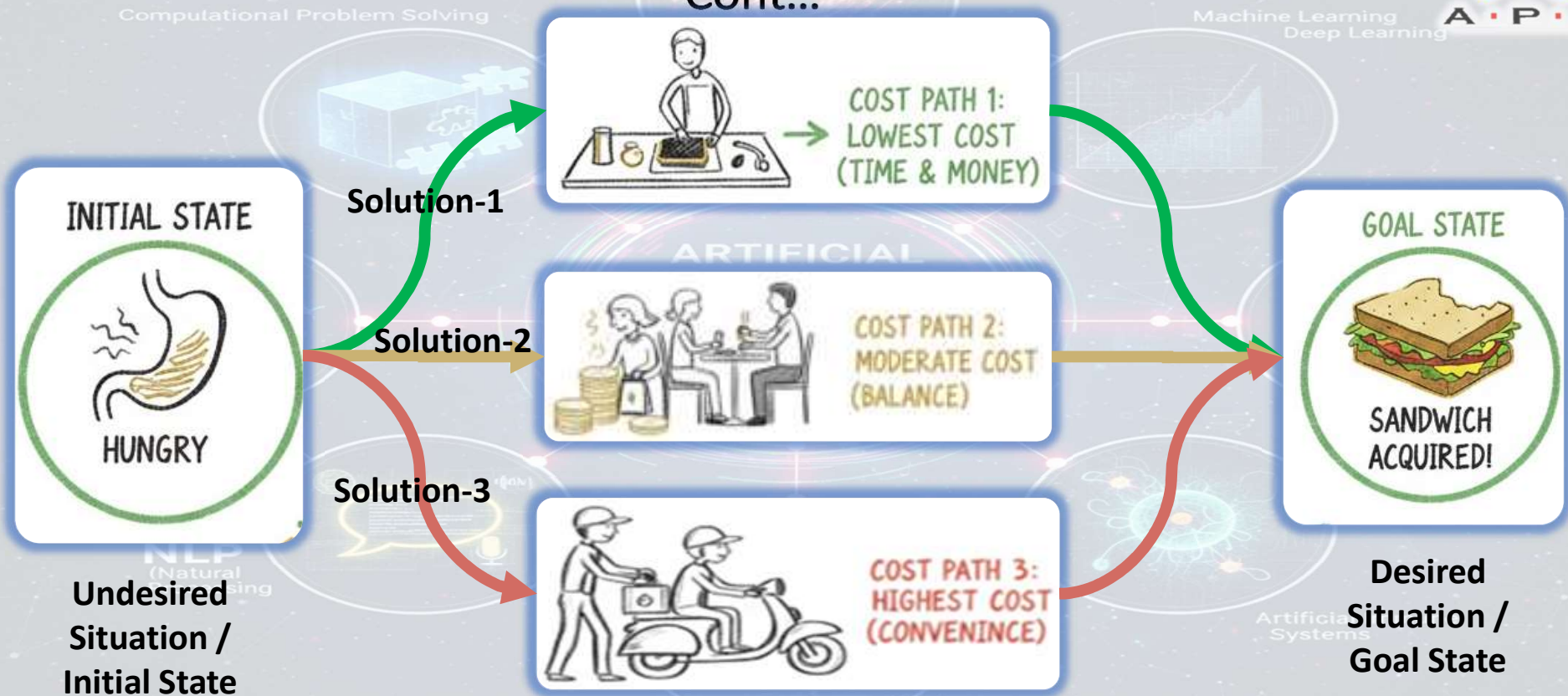


What is a Problem?

A **problem** is a situation where a current state is **undesirable** or **unknown**, and there is a **desire to reach a goal state** through a sequence of actions, but the path to that goal is **not immediately obvious**.



What is a Problem? Cont...



HUMAN BEINGS must decide Which solution is the best?

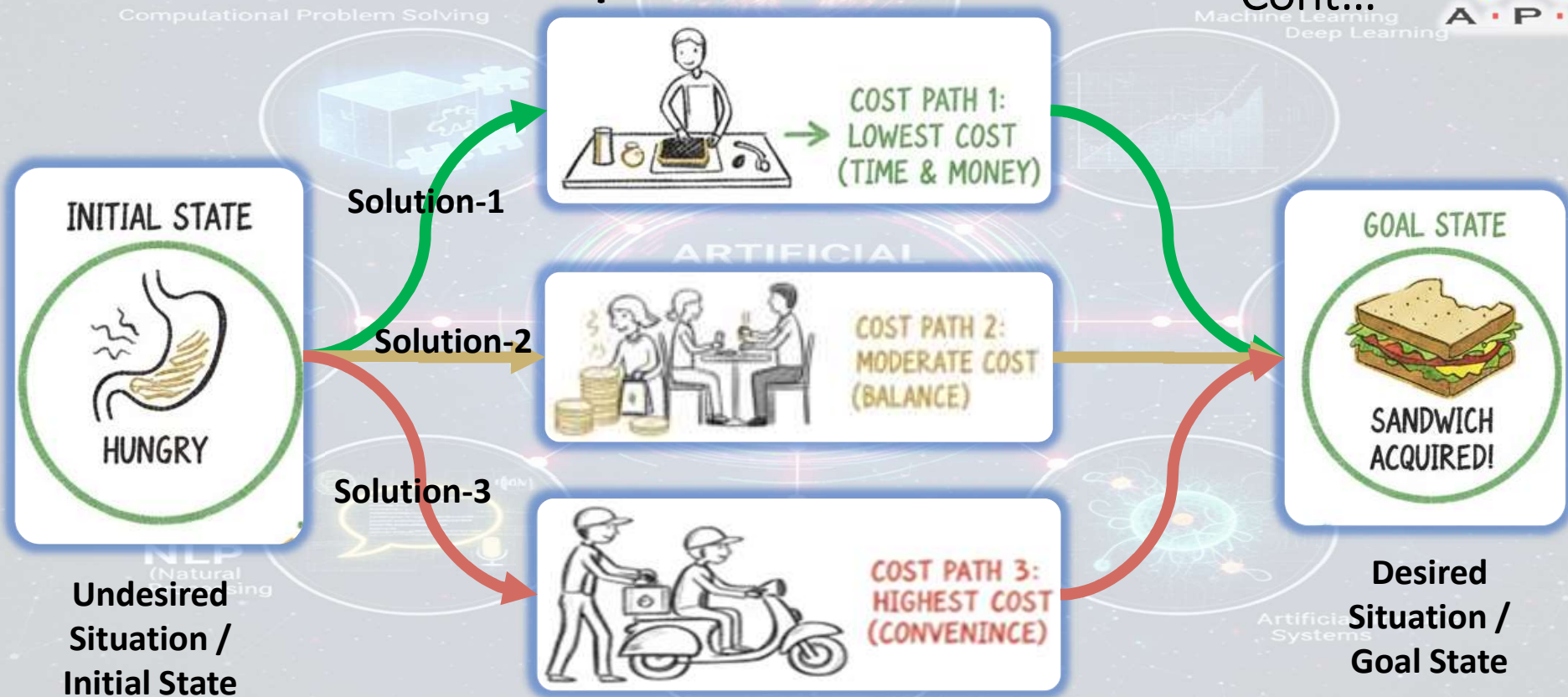


What is an AI Computational Problem?

- An **AI Computational Problem** (or simply an **AI Problem**) is a general problem that has been translated into a **formal, mathematical structure (computerized version)** that an intelligent agent (AI) can manipulate using algorithms.

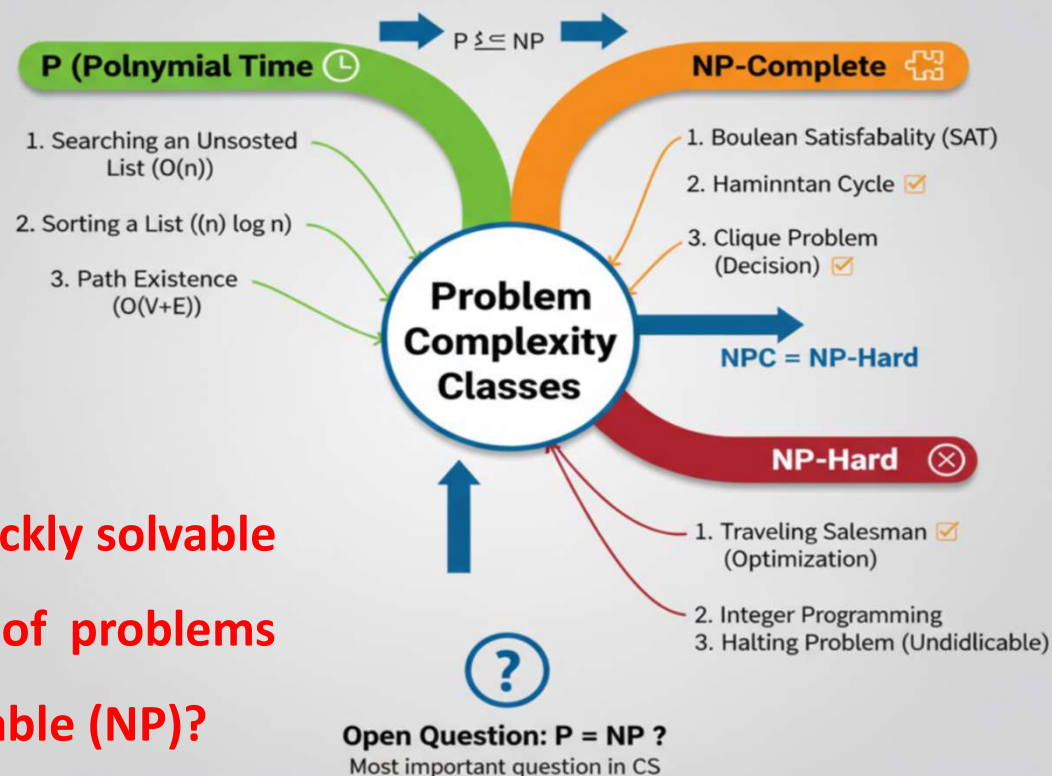


What is a AI Computational Problem? Cont...



AI ALGORITHM will decide Which solution is the best?

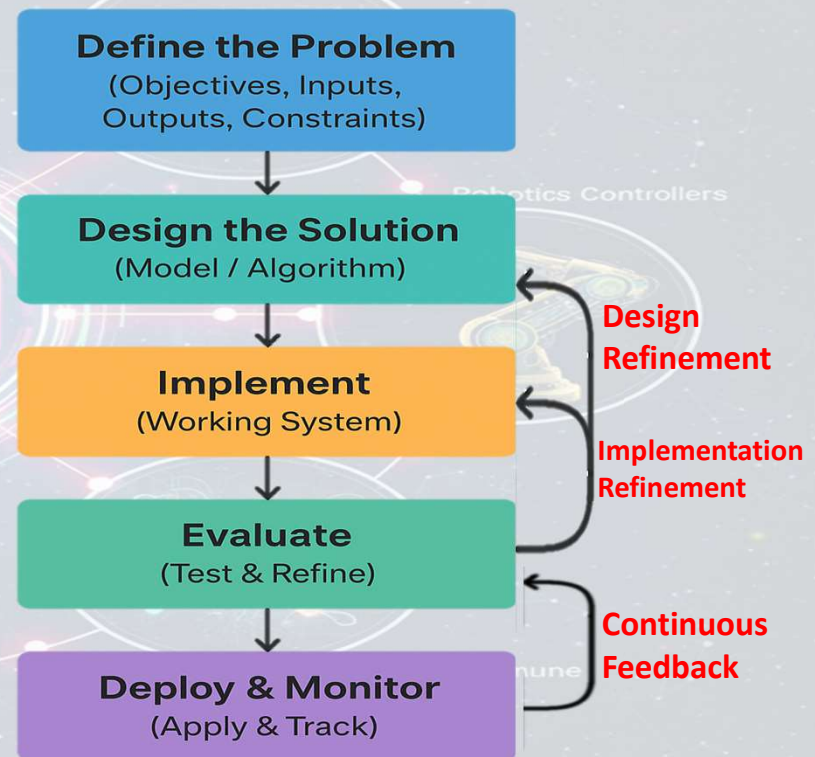
Is the set of problems that are quickly solvable (P) exactly the same as the set of problems whose solutions are quickly verifiable (NP)?





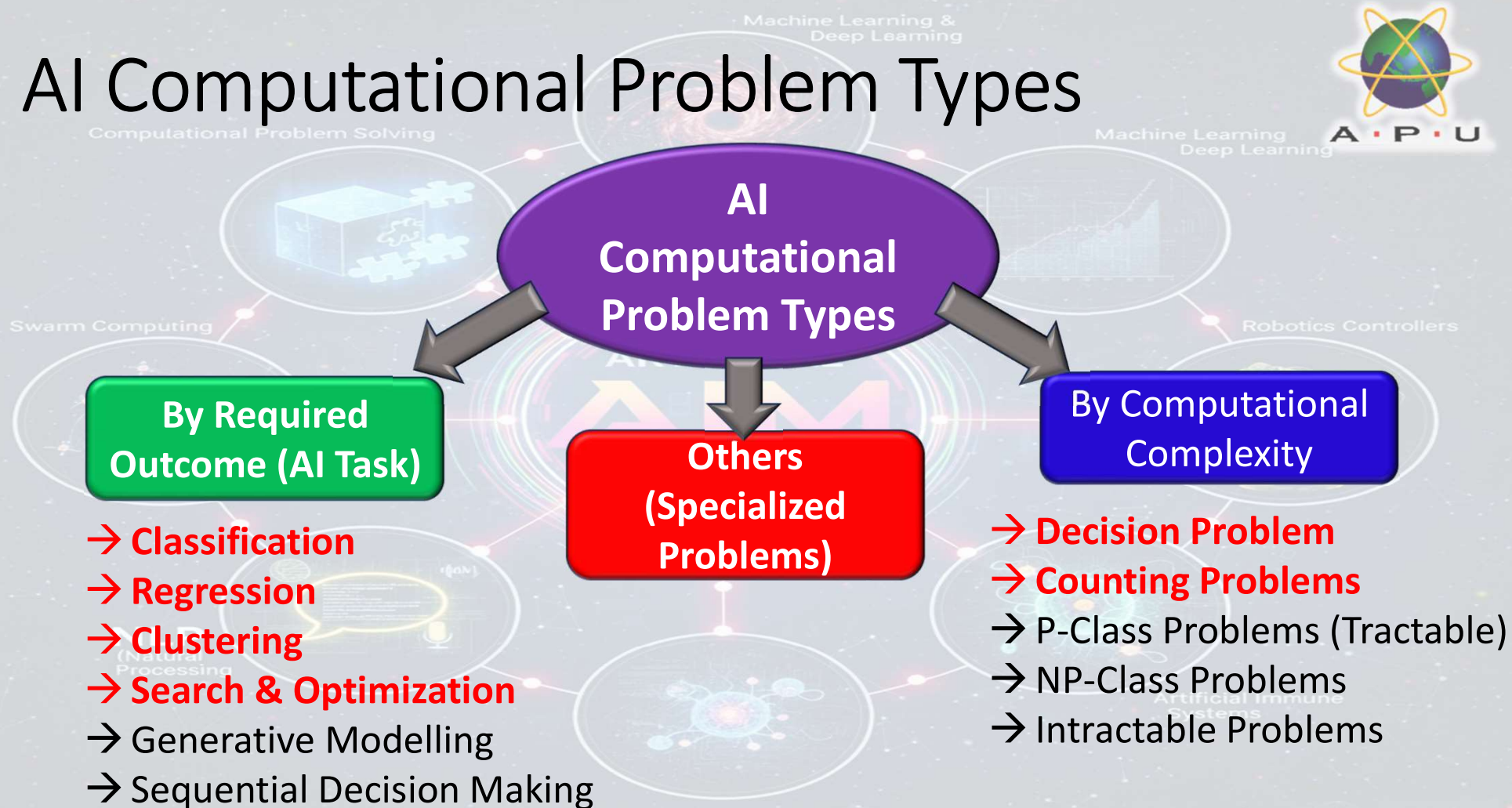
How to Solve Any Problem?

- To solve any AI Computational problem, we need to:
 - **Define and understand the Problem** – Understand objectives, inputs, outputs, and constraints.
 - **Design the Solution** – Choose or develop a suitable model/algorithm.
 - **Implement the Solution** – Translate the model into a working system.
 - **Evaluate** – Test, measure performance, and refine.
 - **Deploy & Monitor** – Apply the solution and track its effectiveness.





AI Computational Problem Types





Decision Problem

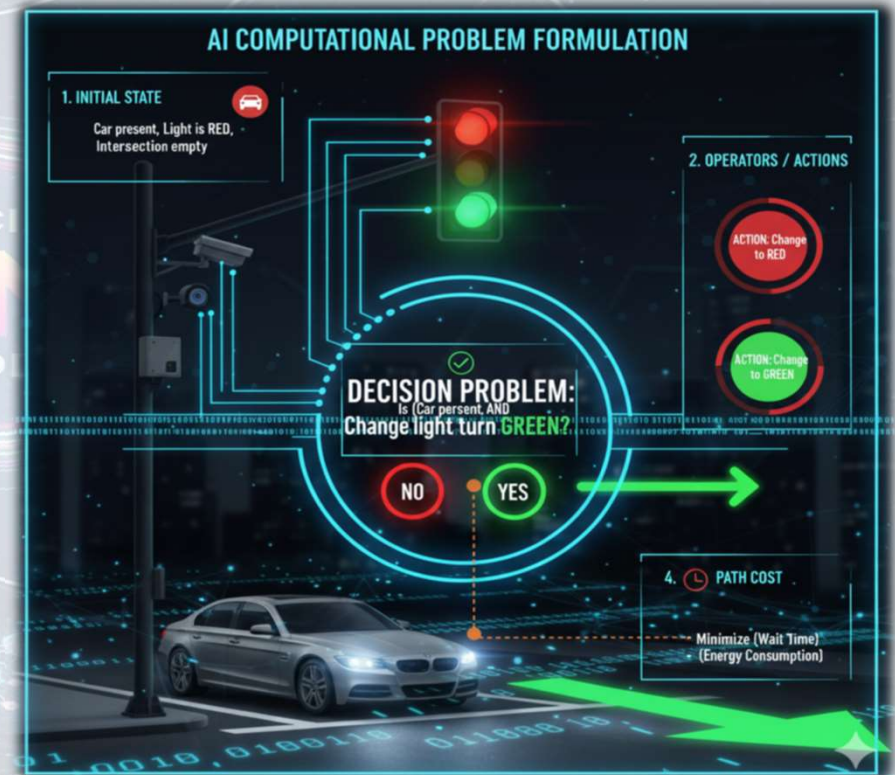
- **A Decision Problem** is any problem that has a simple Yes/No or True/False answer. **DO'S AND DONT'S**
 - **Goal:** To decide whether a given input has a certain property.
 - **Example:** "Is this image a picture of a cat?" (Yes/No)
 - **Example:** "Does a path exist in this graph?" (Yes/No)



Example 1: The Traffic Light (Binary Classification)



- **Input/Current State (Sensor Reading)**
 - Camera detects a car waiting at the intersection.
- **Decision Problem (The Question)**
 - "Should the light change from Red to Green?"
- **Yes/No Answer (Action Taken)**
 - YES (Change to Green) or NO (Stay Red)
- **Goal/Objective (Traffic Flow)**
 - Efficiently move traffic while preventing accidents.





Example 2: The Medical Triage (Risk Assessment)

- **Input/Current State (Sensor Reading)**

- Patient's Status

- **Decision Problem (The Question)**

- "Exhibiting Symptoms of Sepsis?"

- **Yes/No Answer (Action Taken)**

- YES (RED ALERT) or NO (Normal Patient)

- **Goal/Objective (Patient's Safety)**

- Identifying Life-Threatening case early.

**Decision Problem:
The Question**



Is this patient exhibiting symptoms of Sepsis?



**Goal/Objective:
Patient Safety**

Identify life-threatening conditions early.

**Input/Current State:
Patient Vitals**



Is this patient exhibiting symptoms of Sepsis?



**FAST HEART RATE:
FAST & RATE: 110 BPM**

AI Component



**Knowledge Base:
Database of Symptoms**

**Triage Result:
Yes/No Answer**



**YES (Alert ER Team
IMMEDIATELY!!!)**

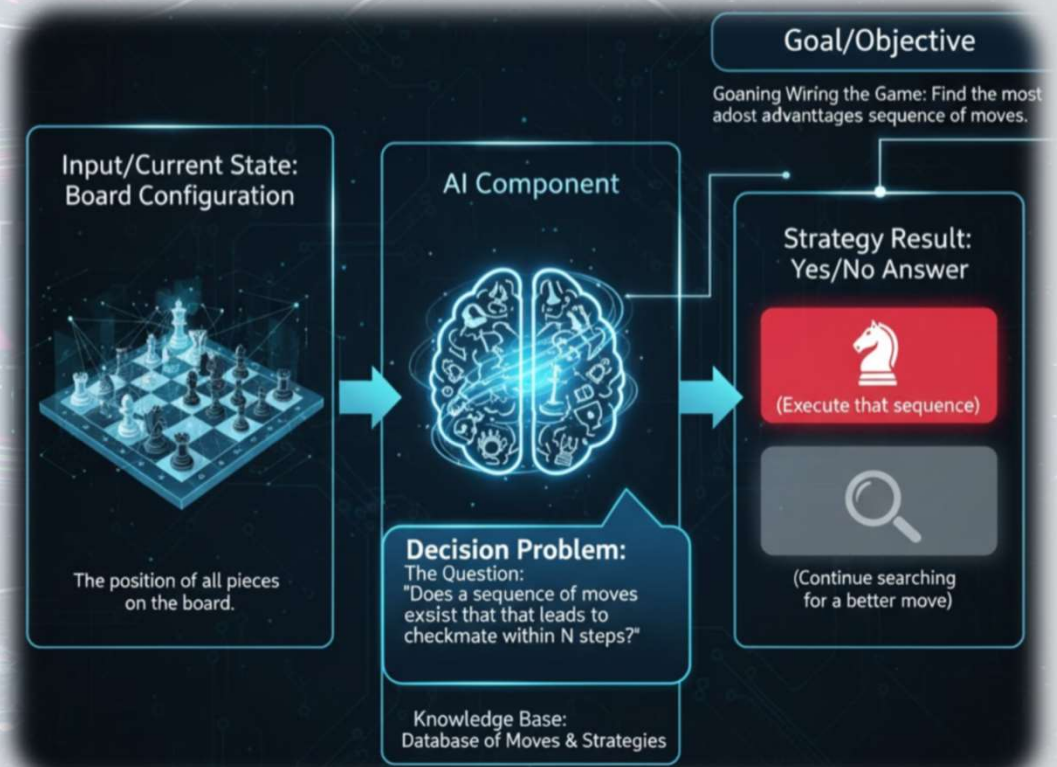
NO

(Monitor as normal)



Example 3: Chess Assistant (IBM's Deep Blue)

- **Input/Current State (Board Config.)**
 - Current Position of Pieces on the board.
- **Decision Problem (The Question)**
 - "Is the following play leading to winning?"
- **Yes/No Answer (Action Taken)**
 - YES (Do the play) or NO (Choose another)
- **Goal/Objective (Winning the Game)**
 - Making sure that we are on the winning way.

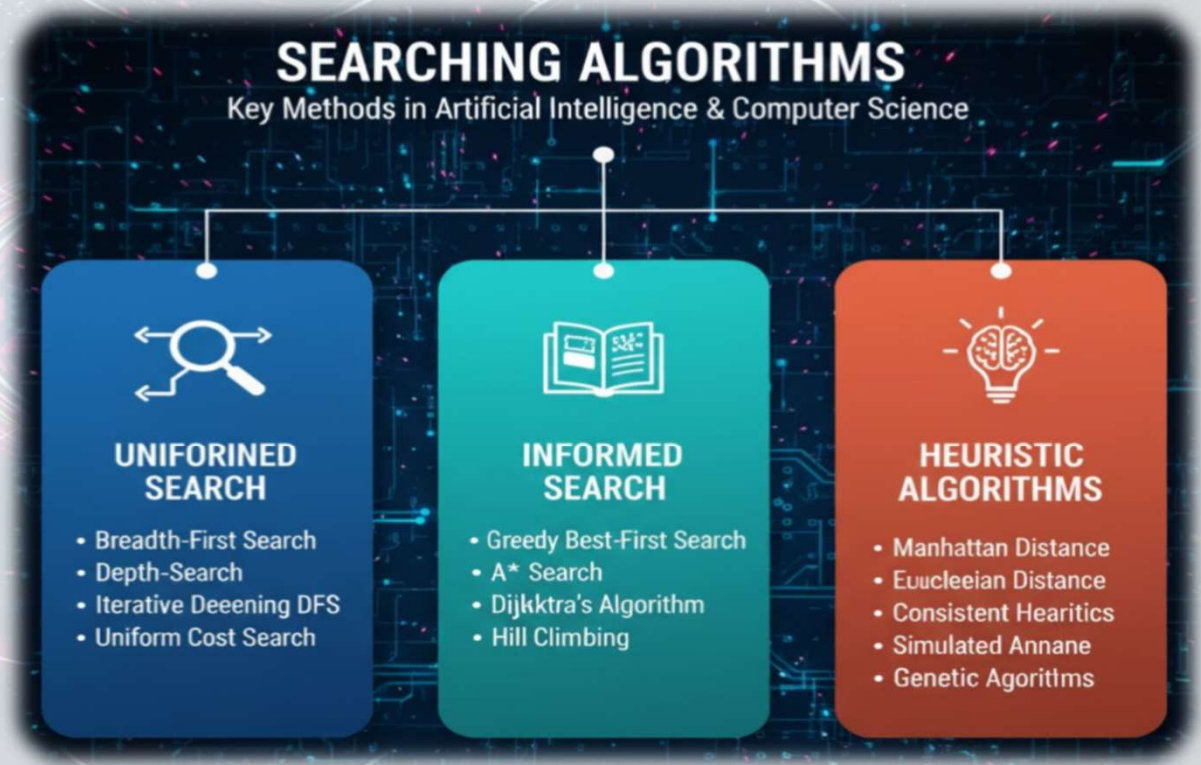


SEARCHING PROBLEMS



Searching Problems

They are the systematic **path-finding strategies** used by an agent to traverse a problem space from an **initial state** to a **goal state**.



Searching Problems State-Space-Diagram



Edge Weights

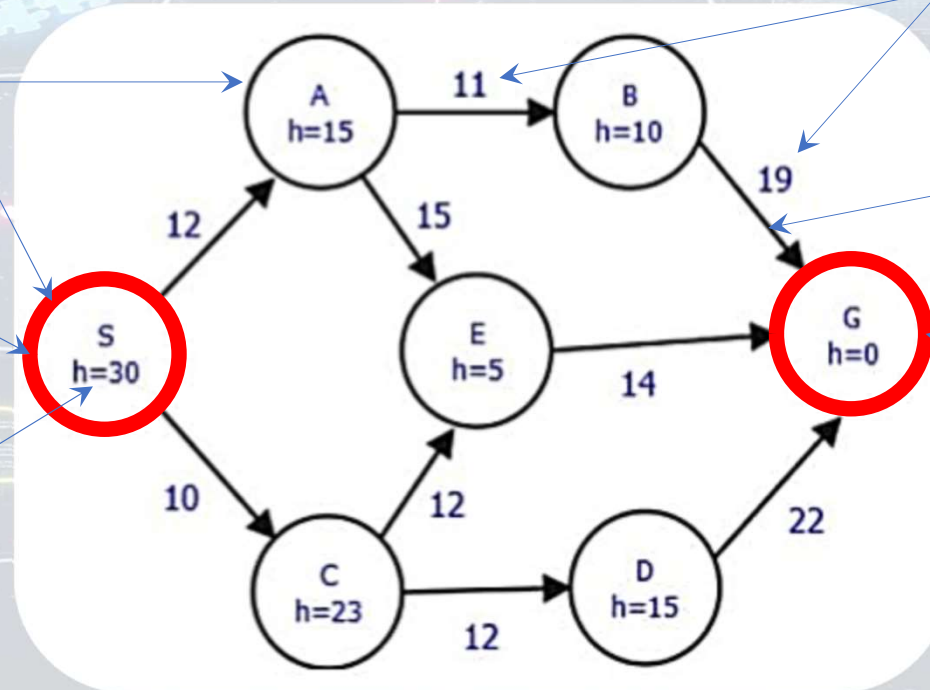
Edges

Goal State

NODES/STATES

Start State

NLP
Heuristics

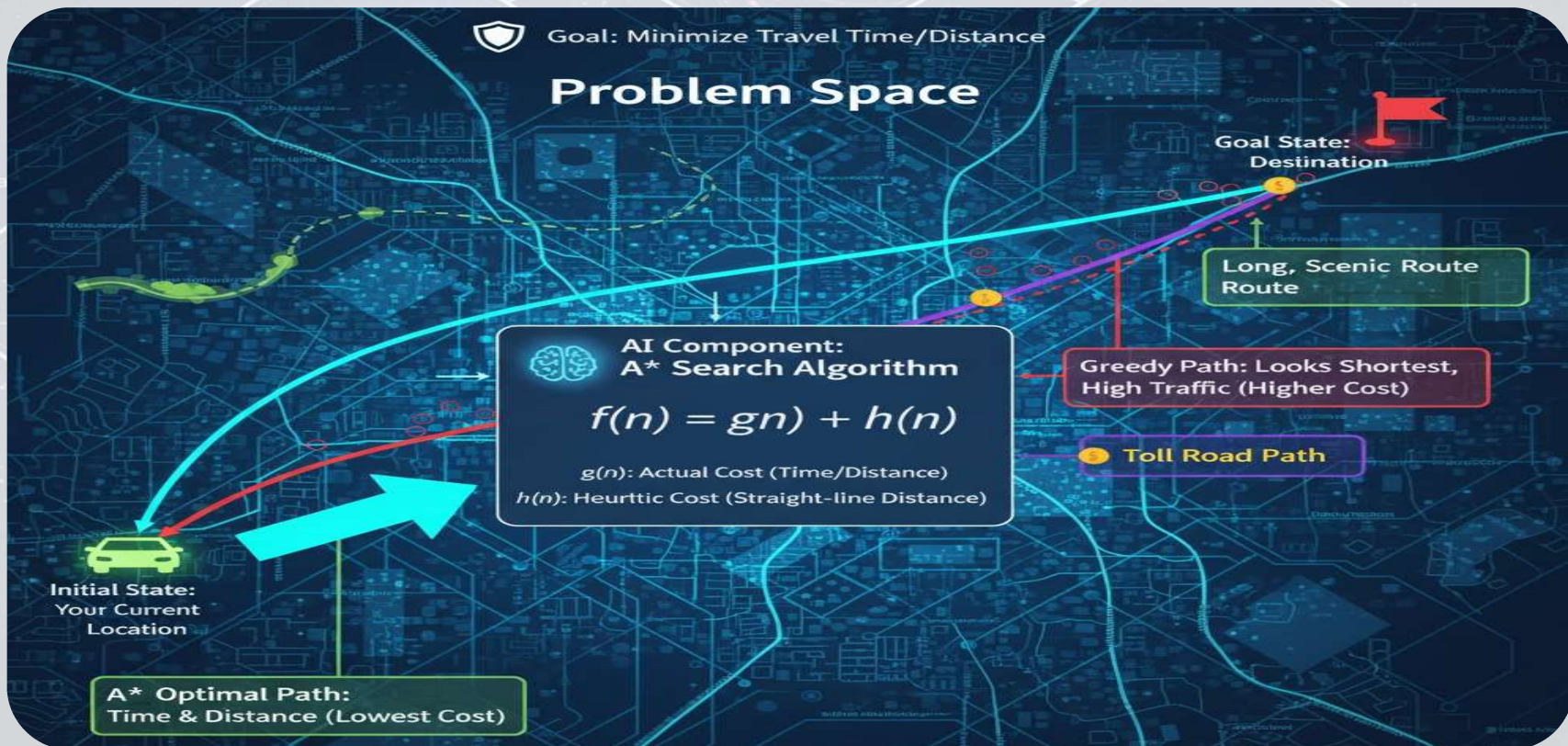




What are Search algorithms?

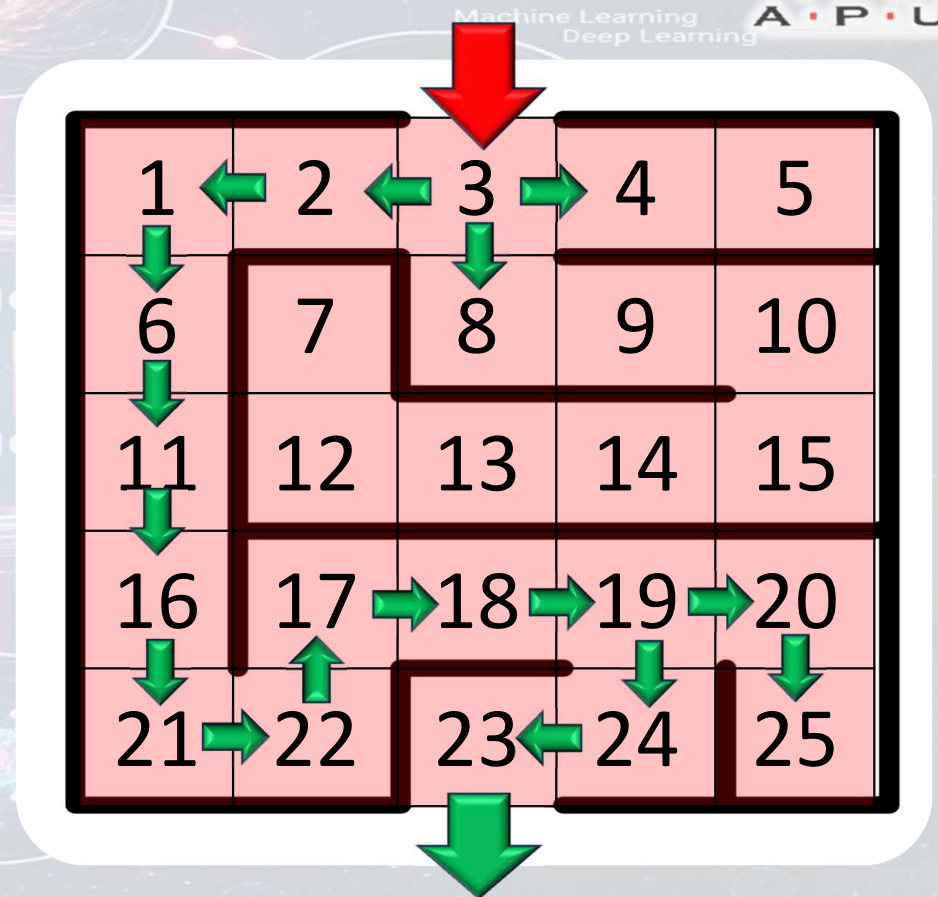
- **Definition:** Search algorithms are methods for finding a path from a starting state to a goal state.
- **Importance in AI:** Used in problem-solving, decision making, and optimizing solutions.
- **Examples of Applications:** Pathfinding in games, puzzle solving, web search, AI planning.

Example 4: GPS Navigation – Optimal Route

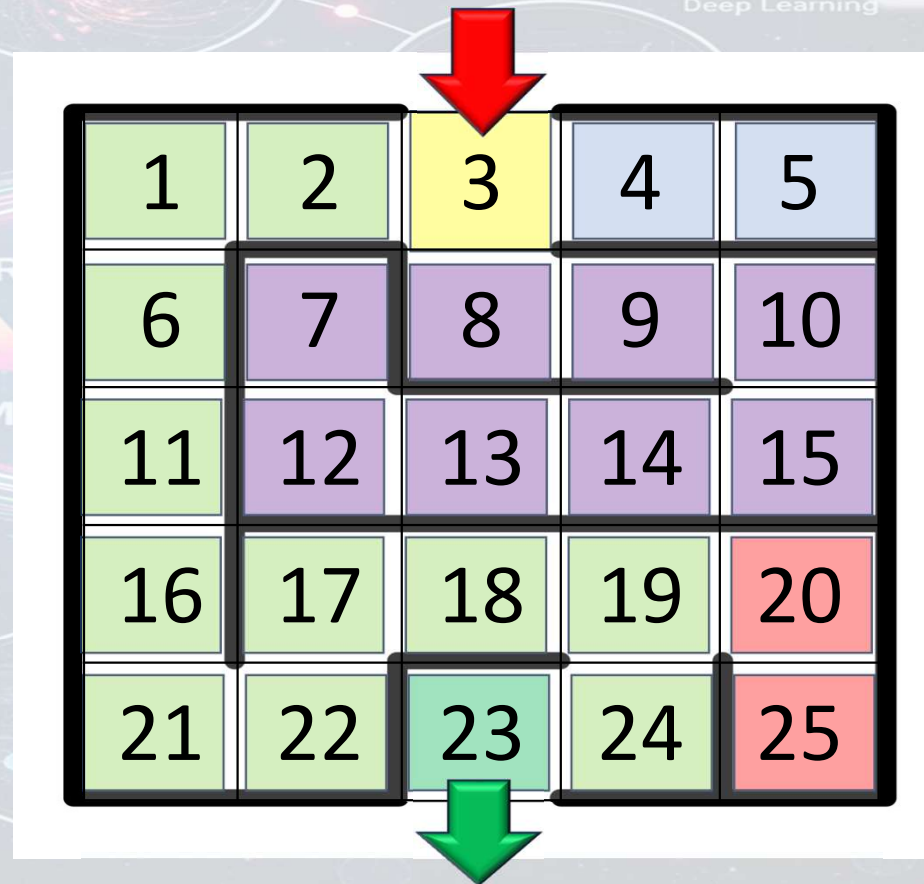
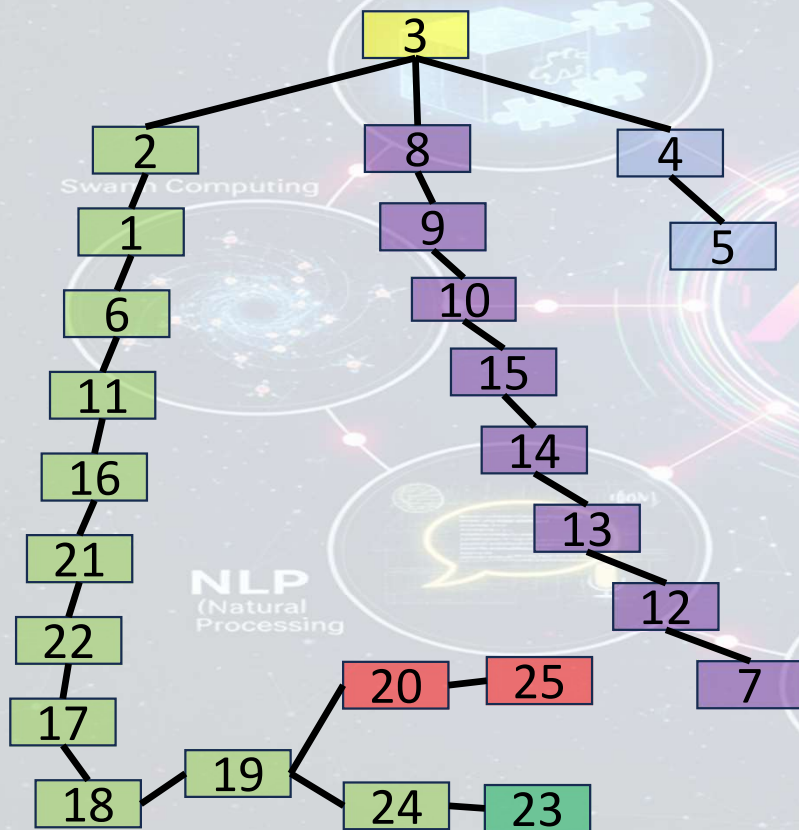


Example 5: Maze Escape

- AI **doesn't "see" walls** as physical barriers; it only processes the grid as a collection of numbered points connected by movement options.
- For example, Cell-3 has three movement-options; 2, 8, and 4.
- On the other hand, Cell-17 has only one movement-option, which is to Cell-18.



Example 5: Maze Escape (as seen by AI)

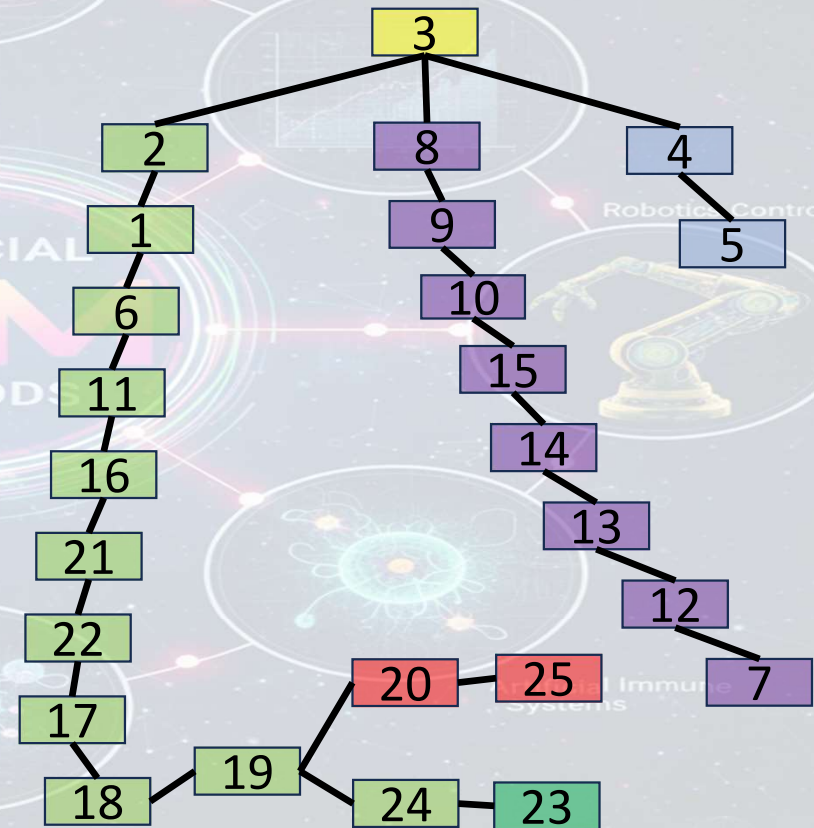




Example 5: Maze Escape Solution

- Now, can use any Searching algorithm to find a path or the best path.
- Try it yourself using:
 - BFS
 - DFS

Which one is better?!



Example 6 (Homework): Alpha-Go

- Analyze the 1.5-hour movie and find out the following:

- Problem Space
- Initial State
- Desired/Goal state
- Searching Algorithm (s) used.

Youtube: AlphaGo - The Movie

Direct Link:

<https://www.youtube.com/watch?v=WXuK6gekU1Y>

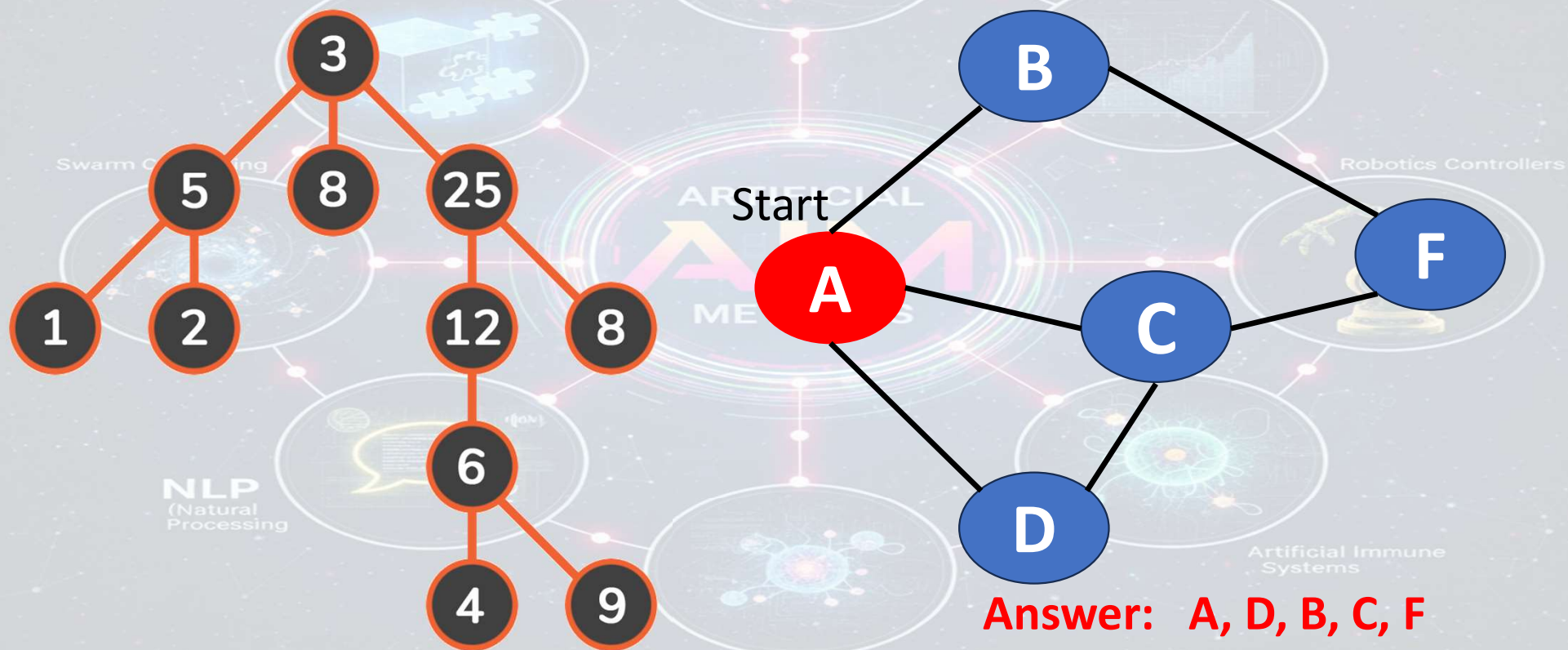




Searching Algorithms Revision & New Algorithms (Untaught in IAI)



Breadth-First Search (BFS)

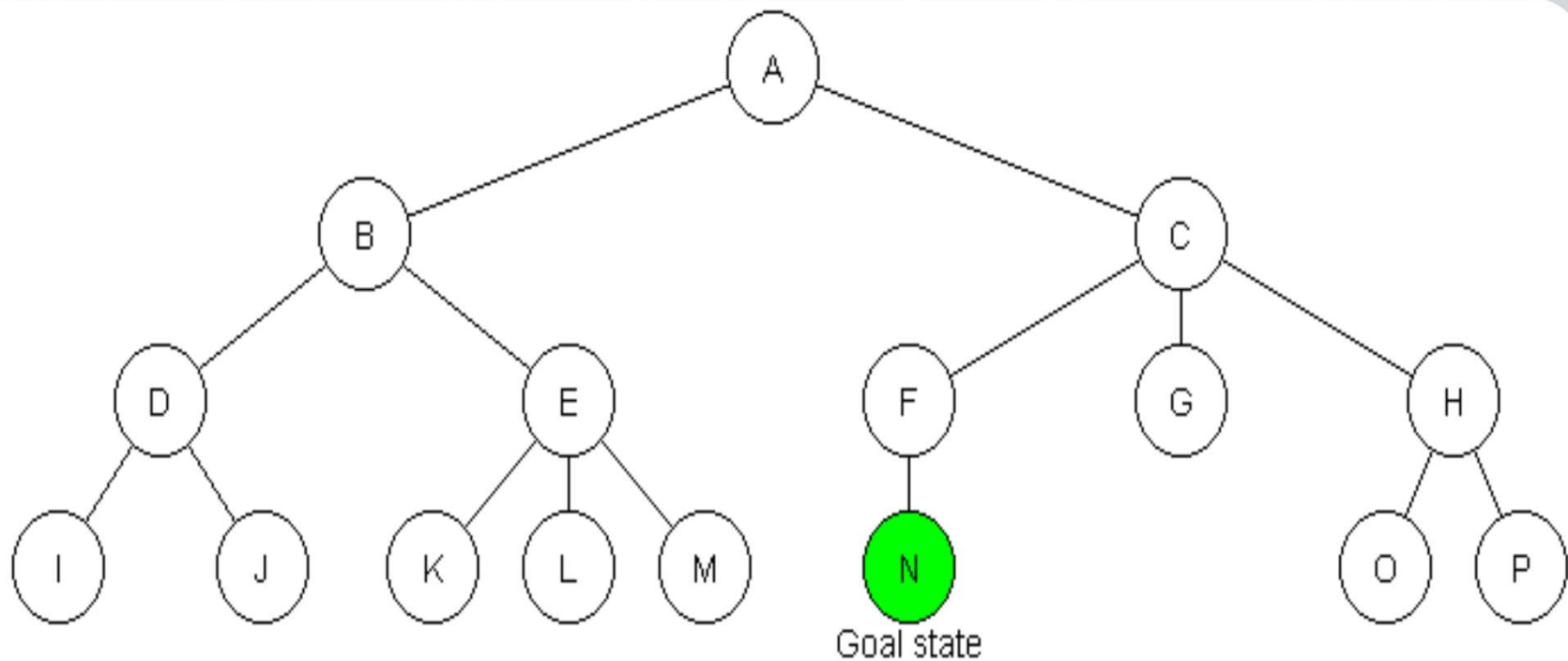


Depth-First Search (DFS)

Computational Problem Solving

Machine Learning &
Deep Learning

Machine Learning
Deep Learning



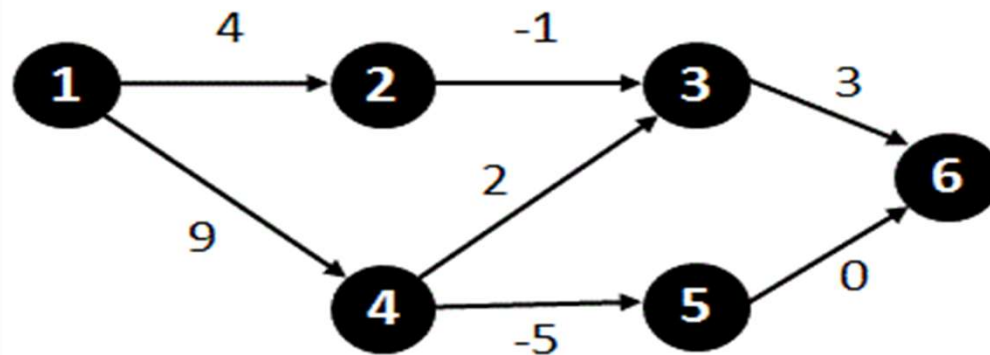
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Goal state

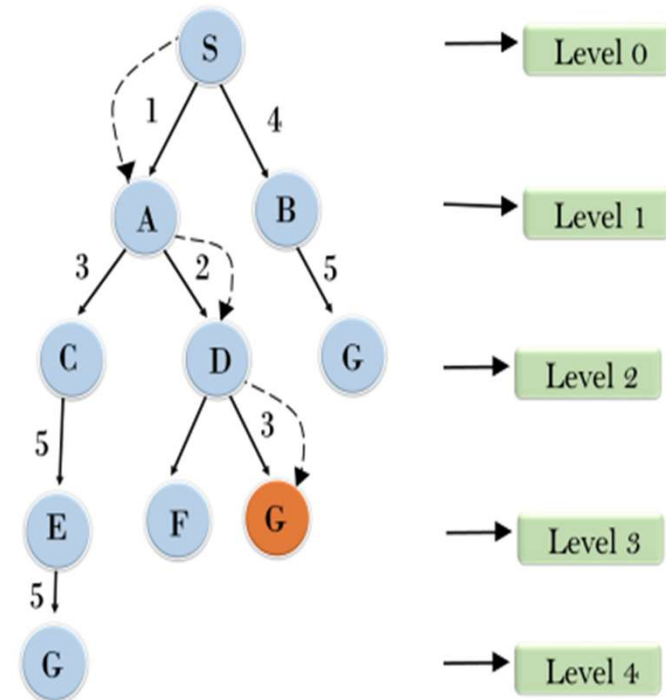
Uniform Cost Search (UCS)

Computational Problem Solving



What is shortest path from 1 to 6 ?

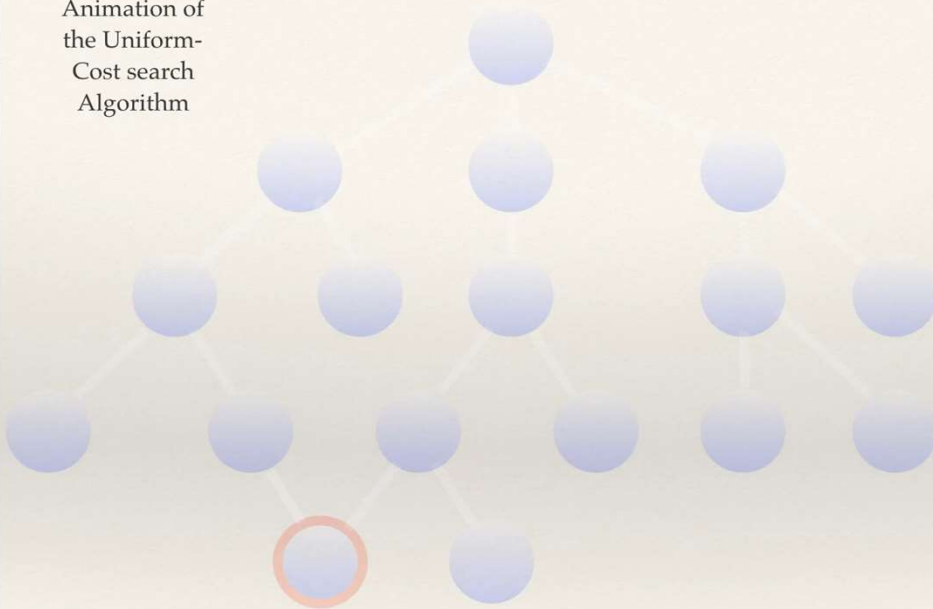
Uniform Cost Search



Uniform Cost Search (UCS)

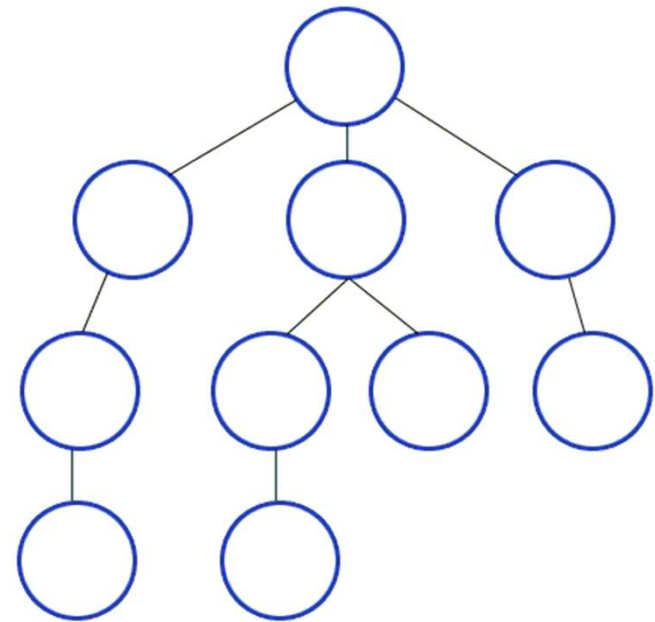


Animation of
the Uniform-
Cost search
Algorithm

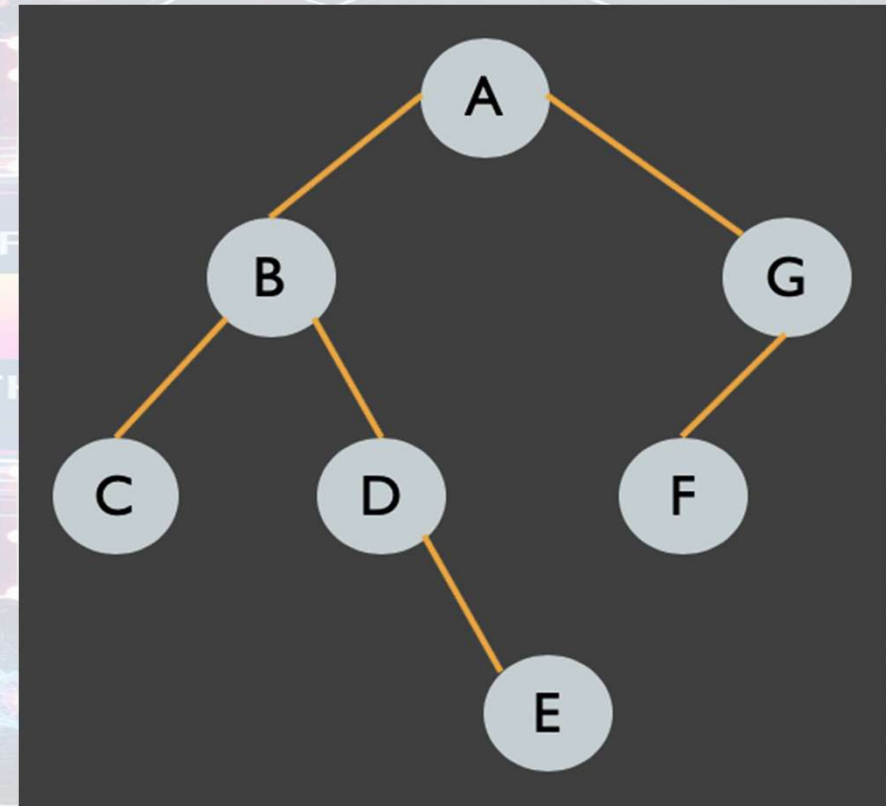
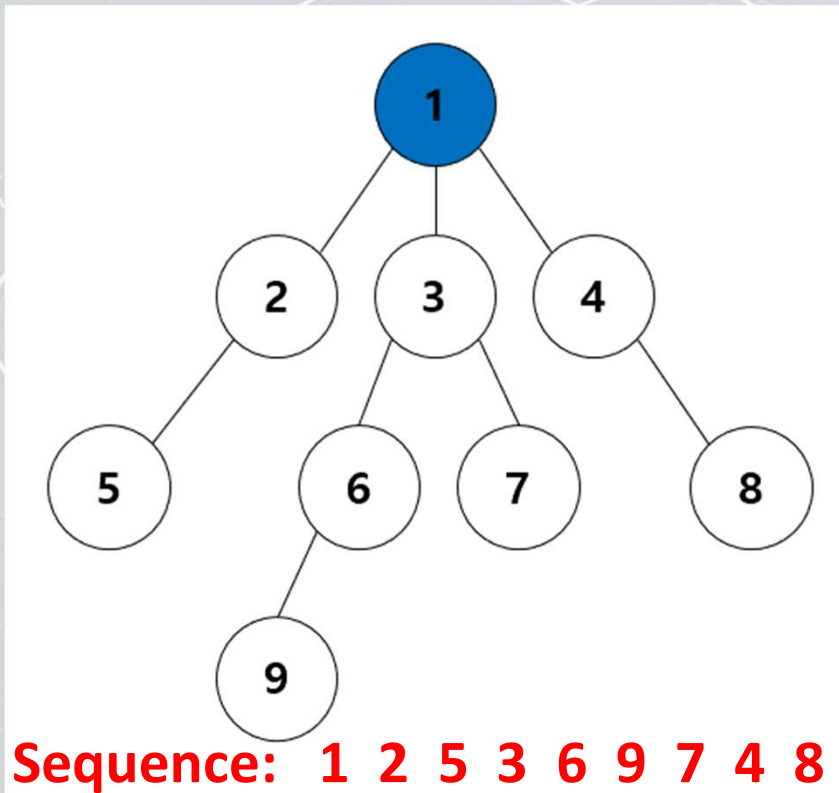


Shaul Markovitch © 2018

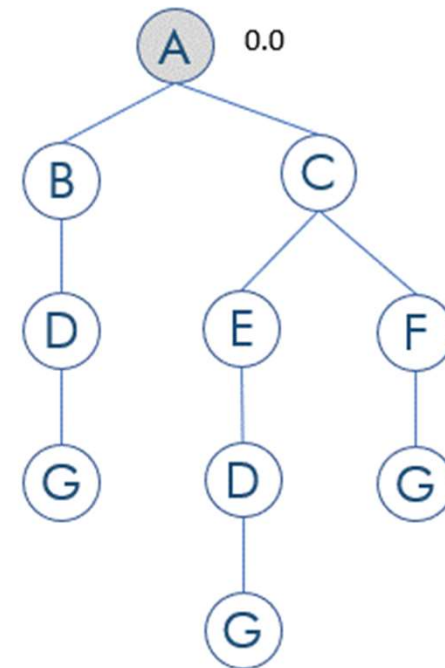
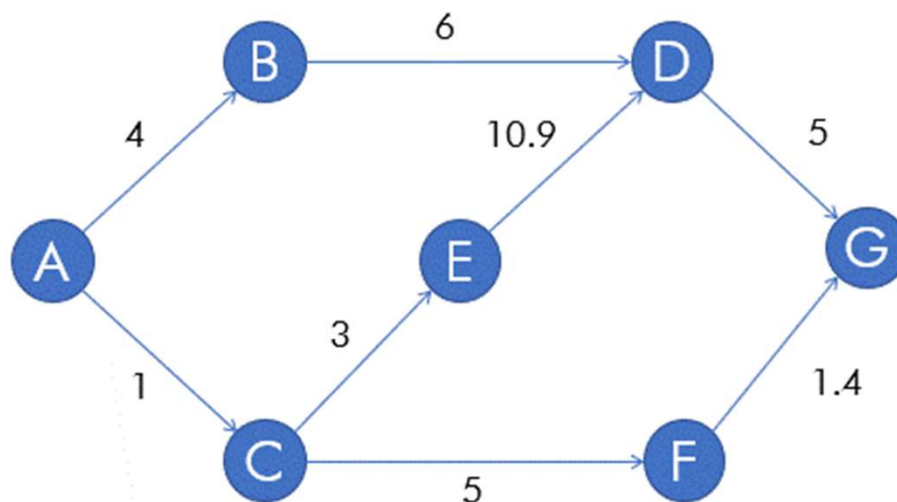
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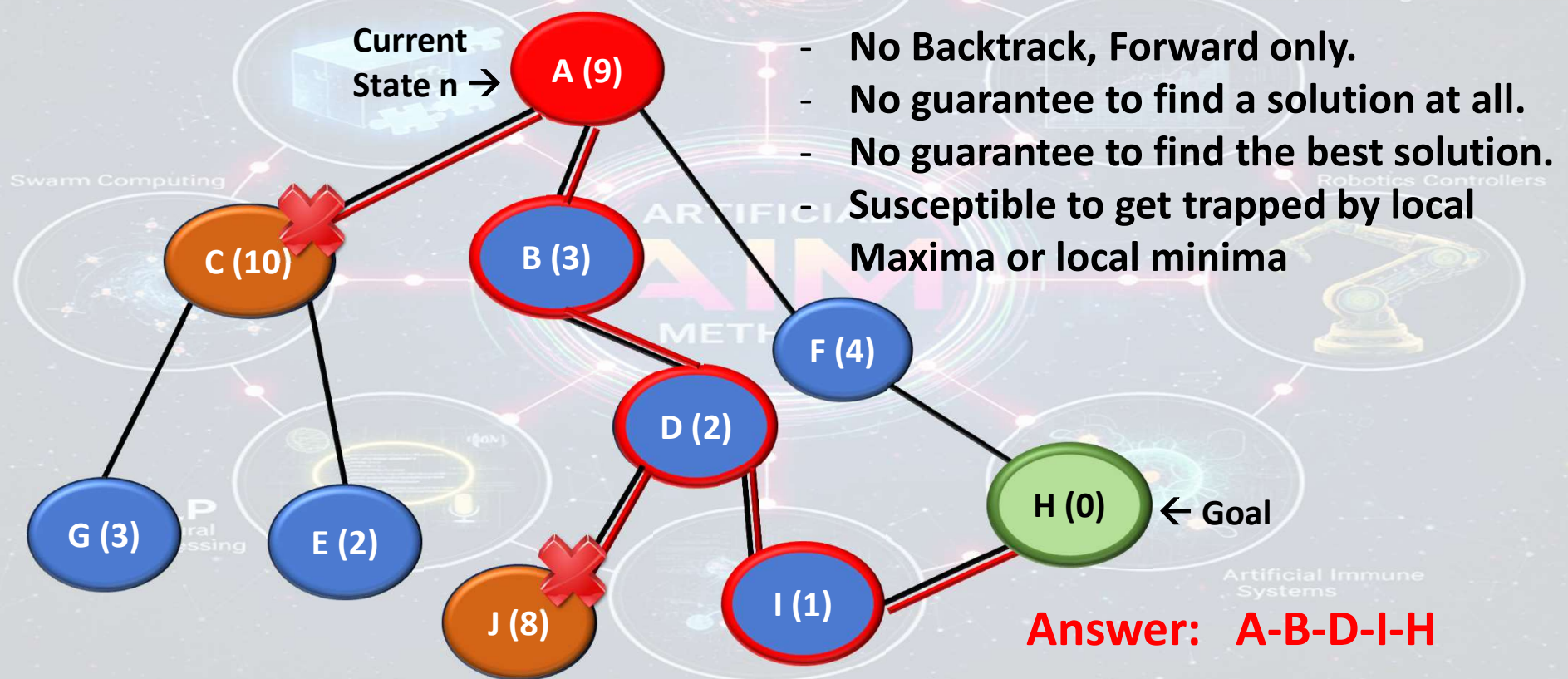
Depth-First Search (DFS)



Uniform Cost Search (UCS)



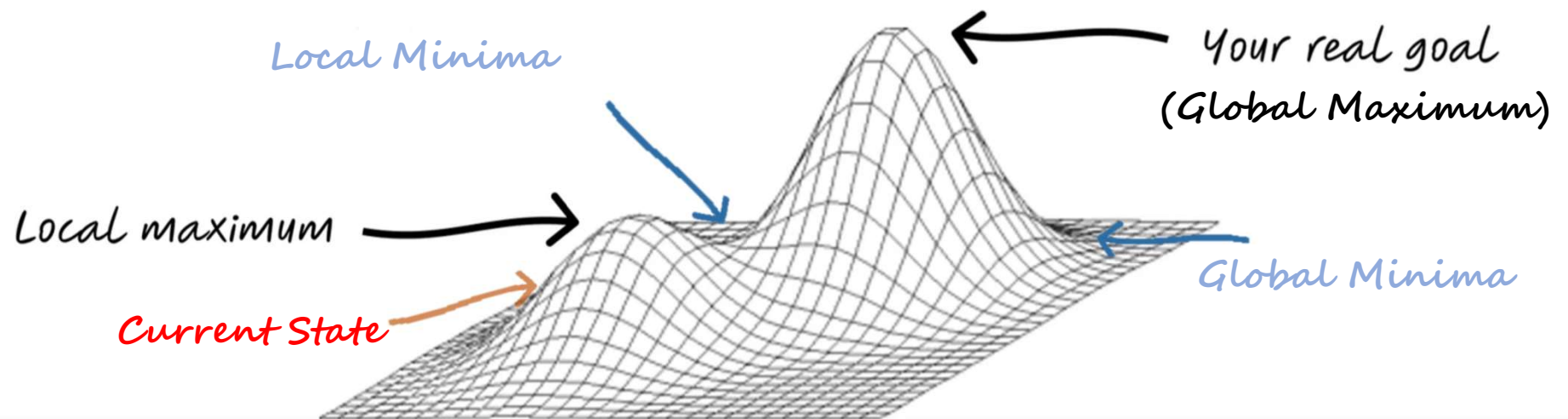
Informed Climbing Hill/Greedy Search



Local Maxima Vs. Global Maxima

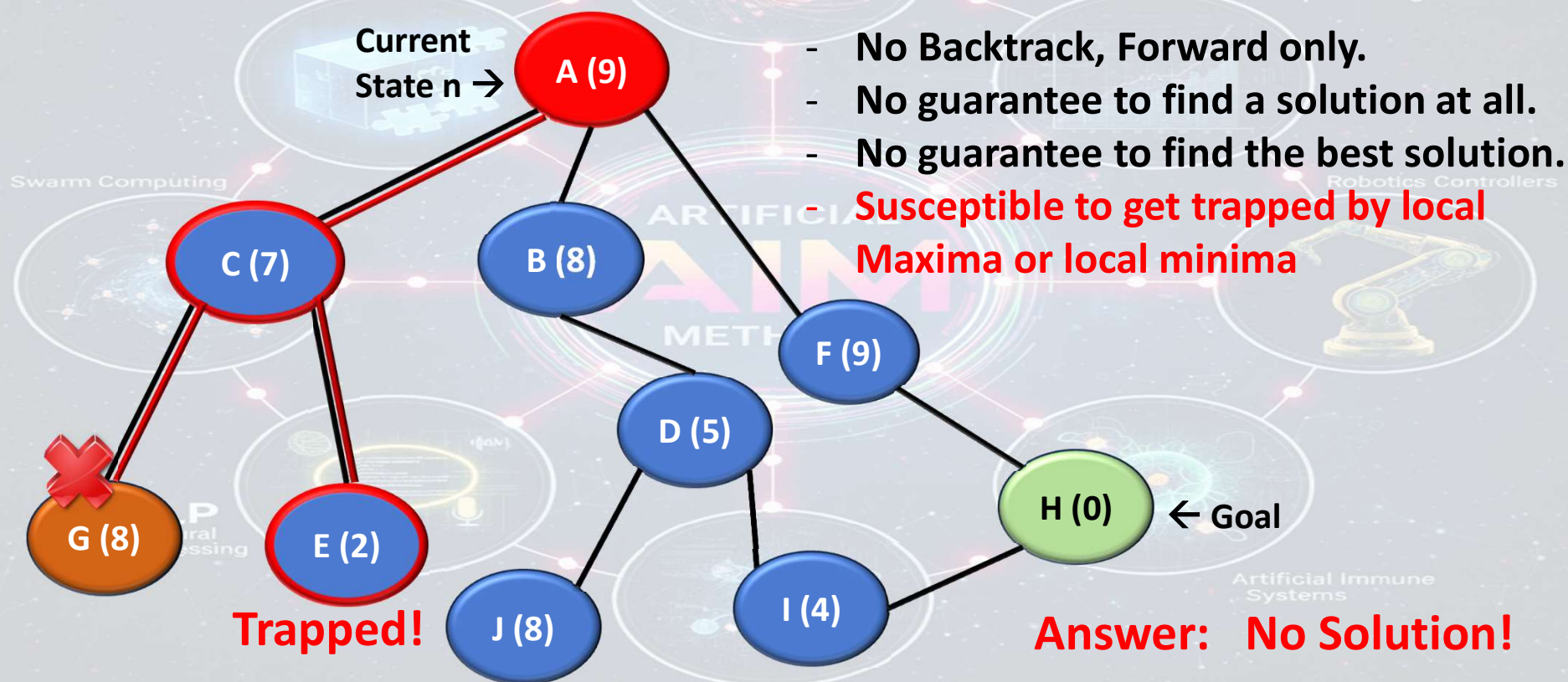


Climbing The Wrong Hill



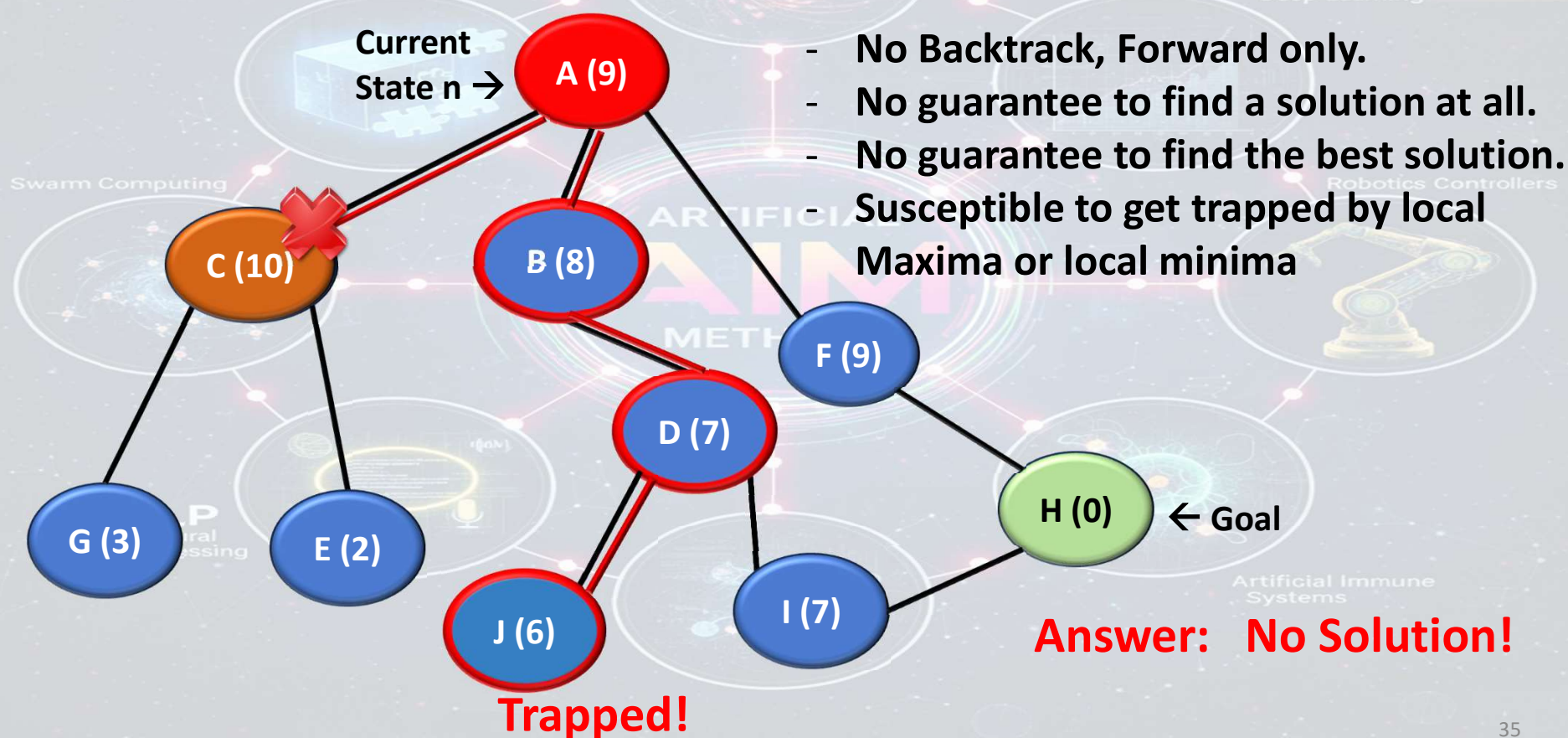


Informed Climbing Hill/Greedy Search



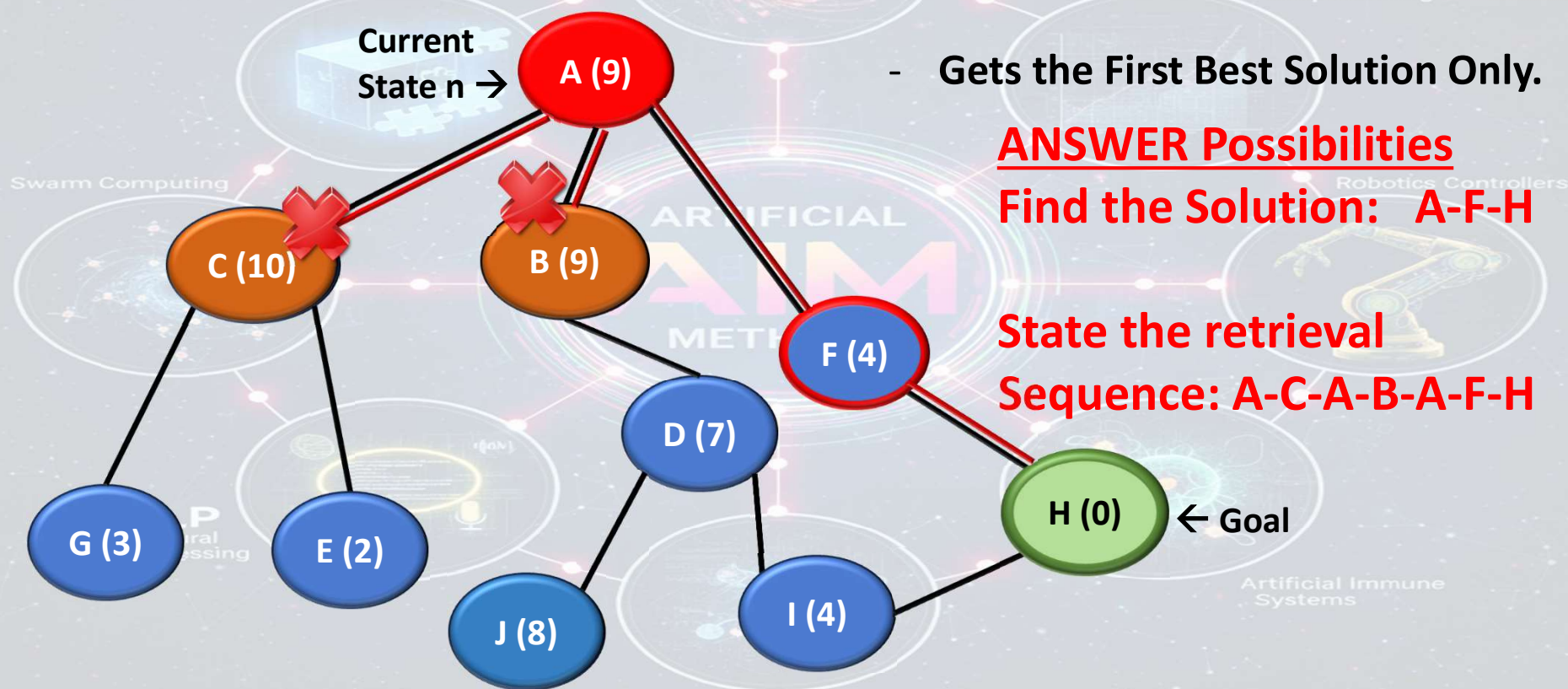


Informed Climbing Hill/Greedy Search





Informed Climbing Hill/Greedy Search

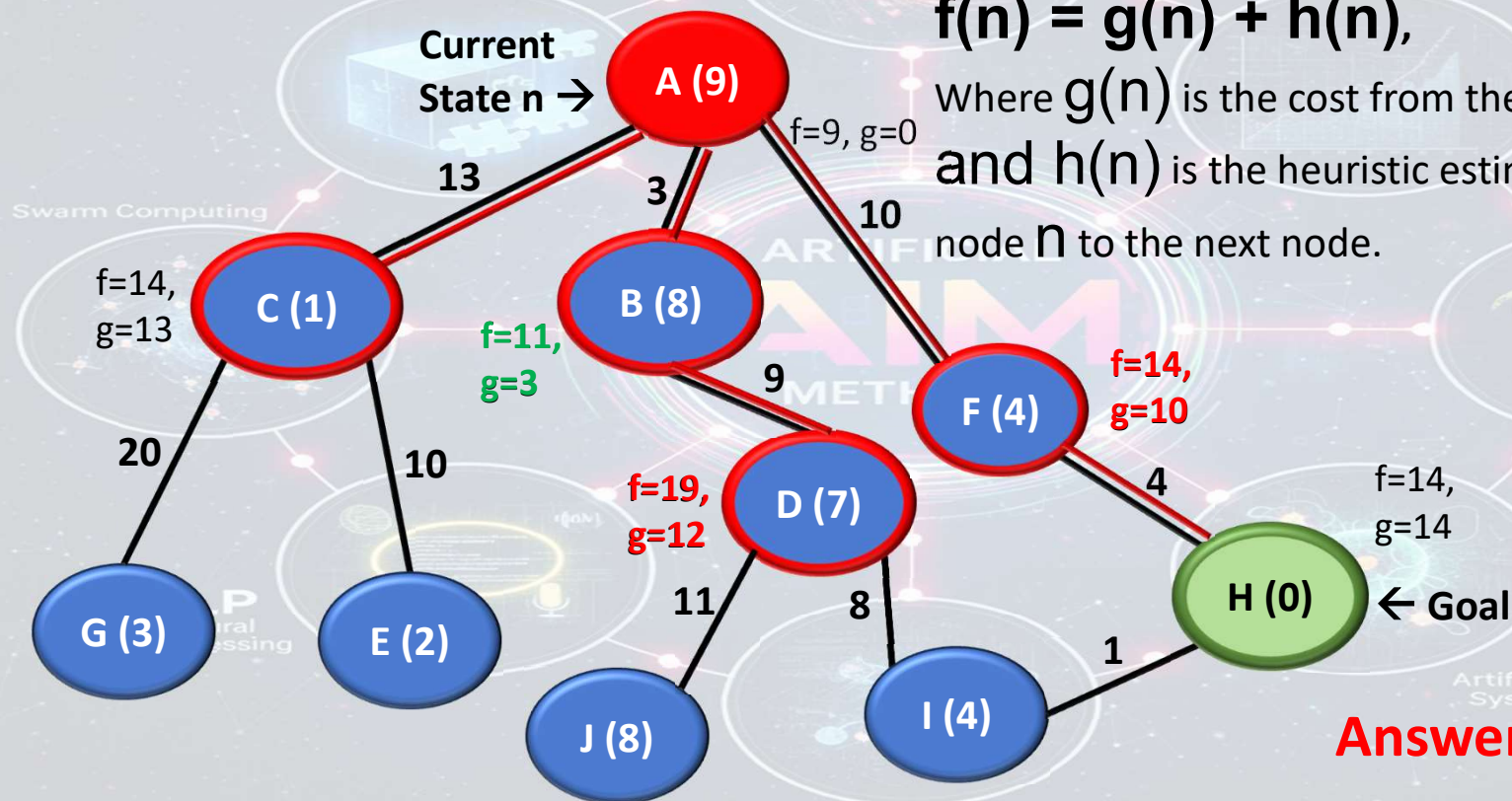


Informed A* Search



$$f(n) = g(n) + h(n),$$

Where $g(n)$ is the cost from the start node to node n , and $h(n)$ is the heuristic estimate of the cost from node n to the next node.





Computational Problems vs Solutions

Computational Problem	Computational Solution-Algorithm(s)	Searching Goal
Searching	Binary Search	Locate a specific element.
Shortest Path	Dijkstra's Algorithm	Find the minimum cost path between two points.
All-Pairs Shortest Path	Floyd-Warshall Algorithm	Find the minimum cost path between all node pairs.
Minimum Spanning Tree (MST)	Prim's Algorithm / Kruskal's Algorithm	Find the minimum total weight structure connecting all vertices.
Chinese Postman Problem (CPP)	Blossom Algorithm (for Matching)	Find the shortest path traversing every edge.
Game Tree Search	Minimax Algorithm (with Alpha-Beta Pruning)	Find the optimal move to maximize agent's score.



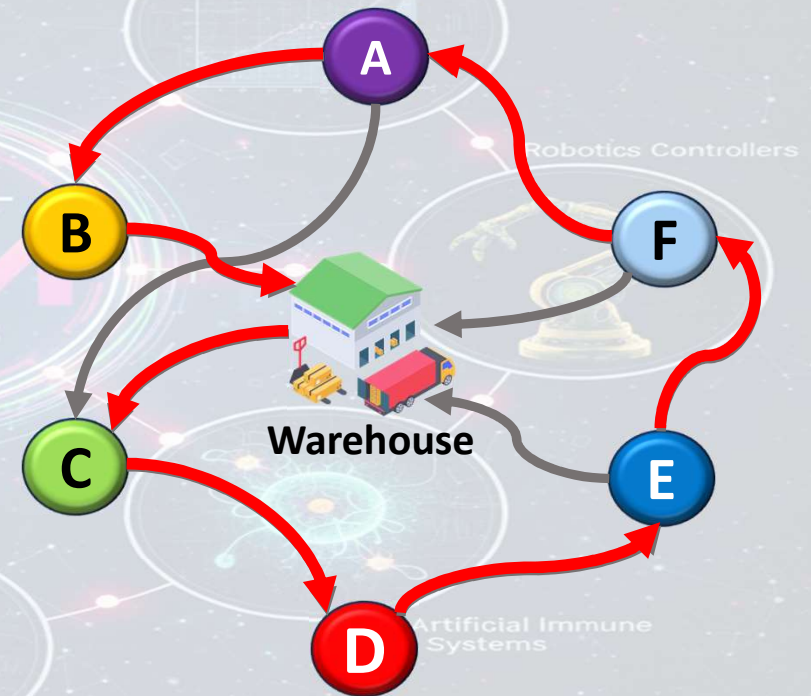
Optimization Problems vs Solutions



Computational Problem	Computational Solution-Algorithm(s)	Optimization Goal
Traveling Salesperson Problem (TSP)	Dynamic Programming (e.g., Held-Karp) / Approximation Algorithms	Minimize the total distance of a route that visits all nodes.
Knapsack Problem	Dynamic Programming	Maximize the total value of items placed in the knapsack without exceeding a weight capacity.
Maximum Flow (Natural Processing)	Ford-Fulkerson Method (via Edmonds-Karp)	Maximize the total flow rate from a source node to a sink node in a network.
Option Pricing	Monte Carlo Method (Financial Modeling)	Estimate or maximize the expected value of a financial instrument (an option).

Example-7: Traveling Salesperson Problem (TSP)

- **Optimization Problem:**
 - What is the shortest possible route that visits every city (or location) exactly once and returns to the starting city?
- **Results** → The Optimal Sequence of a Tour.
- **Why it Matters?**
 - Saves Real-world cost and gains efficiency.
 - Reduce costs and increase sustainability



Example-8: Knapsack Problem

- **Optimization Problem:**
 - You have a knapsack with limited weight capacity and items with weights and values.
- **Question:**
 - Which items maximize total value without exceeding capacity?
- **Result:**
 - A subset of items and their maximum total value.
- **Goal:**
 - Maximize total value within weight limits.



4 COUNTING PROBLEMS

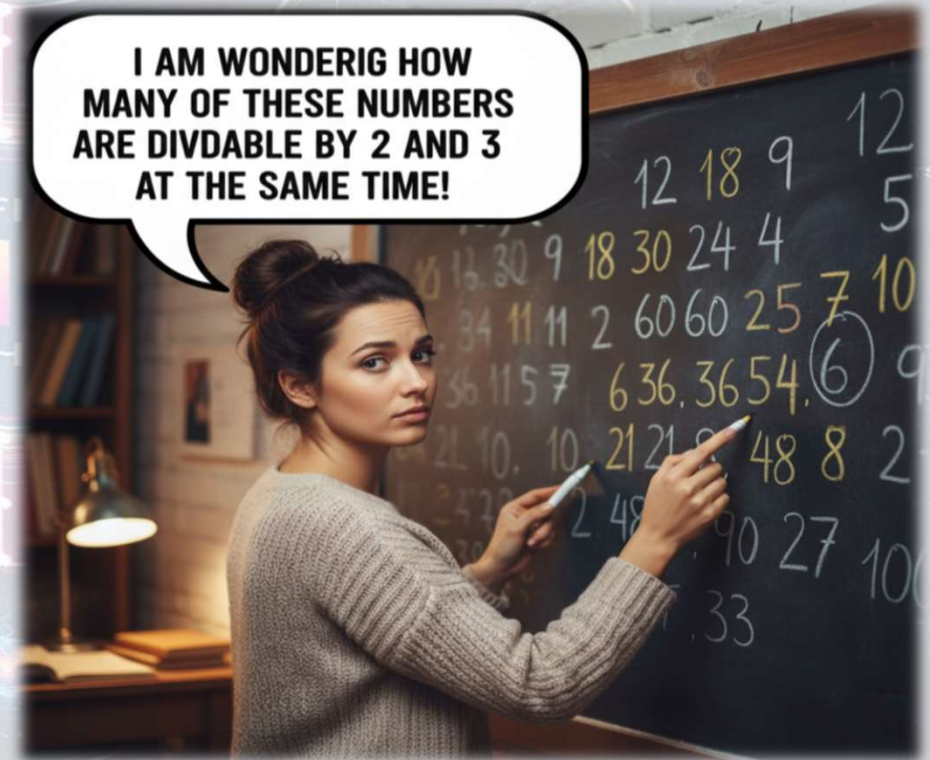




What is Counting Problem?

- **Definition:** An **AI counting problem** (or more formally, a **Counting Problem**) is a type of computational problem whose primary goal is to determine the **number of possible solutions** that satisfy a given set of constraints, rather than finding a single solution itself.

I AM WONDERING HOW
MANY OF THESE NUMBERS
ARE DIVISIBLE BY 2 AND 3
AT THE SAME TIME!





Example-1: Network Reliability and Connectivity

- **Counting Problem:**

- Finding the total number of distinct, valid paths between Node-A and Node-B.

- **Results** → Number of Paths available: 3

- **Why it Matters?**

- To know the network reliability (Higher Path count means a more robust network.)





Example-2: HR's CV-Shortlisting System

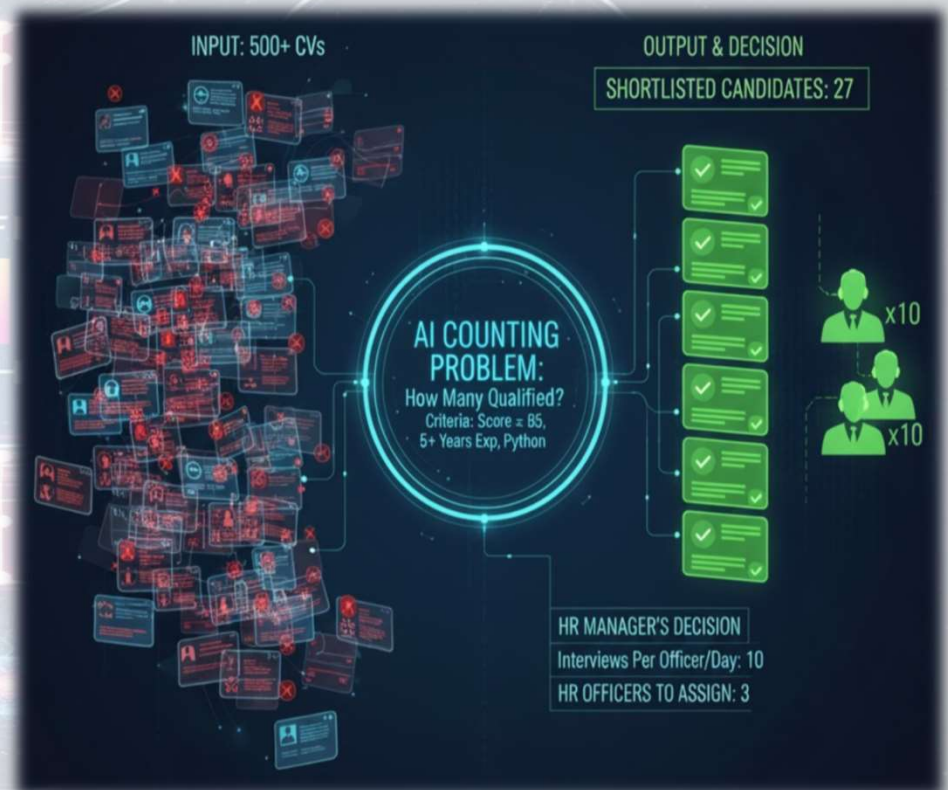
- **Counting Problem:**

- Finding the total number of unique candidate CVs that satisfy the combined minimum score and mandatory criteria.

- **Results** → Number of qualified candidates: 27

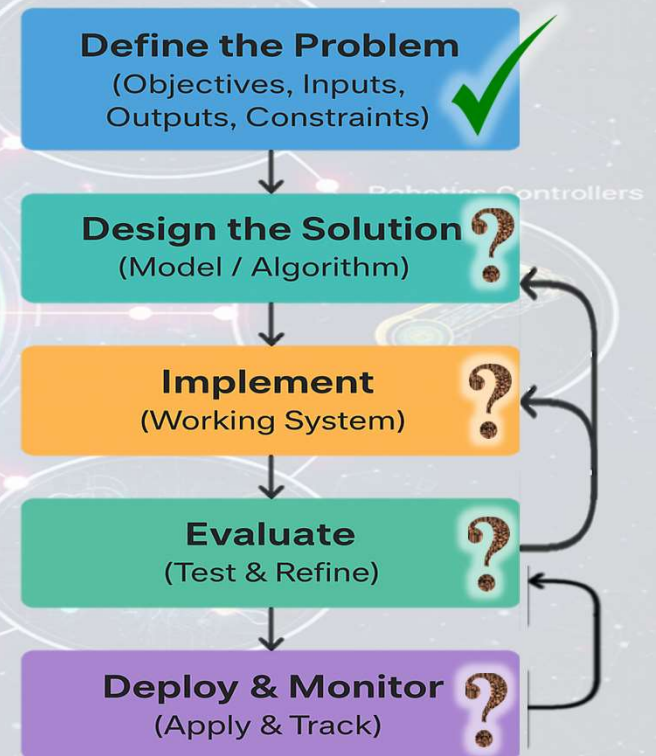
- **Why it Matters?**

- To Allocate a suitable number of interviewers. (If 27 candidates are qualified, then 3 HR Officers are sufficient to process the interviews within a day, as each manages 10 interviews a day at max.)



Wrapping up

- To solve any AI Computational problem, we need to:
 - **Define and understand the Problem** – Understand objectives, inputs, outputs, and constraints.
 - **Design the Solution** – Choose or develop a suitable model/algorithm.
 - **Implement the Solution** – Translate the model into a working system.
 - **Evaluate** – Test, measure performance, and refine.
 - **Deploy & Monitor** – Apply the solution and track its effectiveness.



Wrapping up Cont...1

Computational Problem Solving

Machine Learning &
Deep Learning

Machine Learning
Deep Learning



Problem Type	Short Description
Decision Problem	Is it true or false? Does a solution exist? The output is a simple yes or no answer.
Searching Problem	Where is it? Find a specific item, state, or sequence of actions that leads to a goal.
Optimization Problem	What is the best? Find the single solution that maximizes a desired value or minimizes a cost.
Counting Problem	How many are there? Determine the total number of solutions that satisfy a given condition.
Learning Problem	How can performance be improved? Build a model from data to make accurate predictions or better decisions in the future.

Wrapping up Cont...2



1. An AI system can **solve one or more** computational problems.
2. If an AI system solves more than one computational problem, it's often referred to as a Hybrid AI System (or simply **Hybrid System**).
3. Ultimately, every AI application, from a simple chatbot to a self-driving car, must be fundamentally modeled as a **Decision, Searching, Optimization, Counting, or Learning Problem**.
4. Successfully developing an AI system **begins** with the ability to **correctly identify and categorize the core computational problem** you are trying to solve.



Check Point

25 Questions - 780 Seconds

Question - 1

You want to design an AI system to find out the best way to score in Chess. What kind of computational problem this would be?

- A. Decision Problem
- B. Counting Problem
- C. Searching Problem
- D. Optimization Problem - Is to find the best solution.

Timer  1 Minute



Question - 2

While driving back home using your usual way, you found that an accident led to a terrible traffic jam that might cost you hours to reach home. You used a GPS supported with AI to find out if there is a way to avoid the traffic-jam. What kind of computational problem this would be?

- A. Decision Problem
- B. Counting Problem
- C. Searching Problem
- D. Optimization Problem

Timer  30 Seconds

Question - 3

You work as a delivery man and you would like to deliver a parcel to a specific location, but you are not sure about the address of the destination. So, you used a GPS to assist you finding the place on the map. What kind of computational problem this would be?

- A. Decision Problem
- B. Counting Problem
- C. Searching Problem
- D. Optimization Problem

Timer  30 Seconds



Question – 4

You work for the police force, and you are trying to figure out where a criminal person would be by tracing his/her previous steps using an AI system. What kind of computational problem this would be? (You may pick more than one, if applicable.)

- A. Decision Problem
- B. Counting Problem
- C. Searching Problem
- D. Optimization Problem

Timer



30 Seconds

Question - 5

You work for the police force, and you are trying to figure out **the number of possibilities** where a criminal person would be by tracing his/her previous steps using an AI system. What kind of computational problem this would be? (You may pick more than one, if applicable.)

- A. Decision Problem
- B. Counting Problem**
- C. Searching Problem
- D. Optimization Problem

Timer



30 Seconds



Question - 6

You work for the police force, and you are trying to figure out **the list of possible locations** where a criminal person would be by tracing his/her previous steps using an AI system. What kind of computational problem this would be? (You may pick more than one, if applicable.)

- A. Decision Problem
- B. Counting Problem
- C. Searching Problem**
- D. Optimization Problem

- Searching the possible places
- Not to select the most probably one

Timer  30 Seconds



Question - 7

You work for the police force, and you are trying to figure out **the list of possible locations and the most likely place of** where a criminal person would be by tracing his/her previous steps using an AI system. What kind of computational problem this would be? (You may pick more than one, if applicable.)

- A. Decision Problem
- B. Counting Problem
- C. Searching Problem - list of possible location
- D. Optimization Problem - most likely place

Timer  30 Seconds



Question - 8

You work for the police force, and you are trying to **determine the most likely location** of a criminal by analyzing their previous movements using an AI system. Your goal is to find the **single location** that **maximizes the probability** of the criminal being there. What kind of computational problem is this? (You may pick more than one, if applicable.)

- A. Decision Problem
- B. Counting Problem
- C. Searching Problem
- D. Optimization Problem

Timer  30 Seconds



Question - 9

You work for the police force and are analyzing the previous movements of a criminal using an AI system. There are three locations under consideration: A, B, and C. You want to know if location A is the most likely location of the criminal (i.e., does it have a probability strictly higher than locations B and C)? What kind of computational problem is this?

- A. Decision Problem - among selections, pick one only
- B. Counting Problem
- C. Searching Problem
- D. Optimization Problem

Timer  30 Seconds



Question - 10

Which of the following is not an AI Computational Problem?

- A. Deciding which leaf is healthy.
- B. Finding the ripest fruit.
- C. Deciding the gender of unborn child and his hair color.
- D. Deciding whether an auto-driving car to go left or right of the road.

Timer



30 Seconds



Question - 11

The Traveling Salesperson Problem (TSP), which seeks the shortest route visiting all nodes, is a classic example of what class of computational problems?

- A. Probabilistically Bounded-error Polynomial (BPP)
- B. P-complete
- C. Linearly solvable
- D. NP-complete/NP-hard

Timer  30 Seconds



Question - 12

In informed search algorithms like A*, what is the primary role of the heuristic function $h(n)$?

- A. To track the total number of moves made in the game.
- B. To calculate the exact cost from the start node S to node N.
- C. To estimate the cost from the current node N to the goal node G.
- D. To ensure the algorithm explores every node in the search space.

Timer



30 Seconds

Question - 13

In the context of AI search problems, what is the state space?

- A. The time complexity required to solve the problem.
- B. The set of all possible moves available from any single position.
- C. The optimal path found by a search algorithm.
- D. The total set of all possible configurations or situations the system can be in.

Timer



30 Seconds

Question - 14

Unlike Breadth-First Search (BFS), which explores layer by layer, Depth-First Search (DFS) has a potential drawback when traversing an infinite or very deep search space. What is this primary risk?

- A. Its time complexity is always polynomial.
- B. It requires an excessive amount of memory.
- C. It may get trapped and not terminate.
- D. It is not guaranteed to find the shortest path.

DFS will loop infinitely,
thus may not terminate.

Timer  30 Seconds

Question - 15

Fill the gap with a suitable answer:

Breadth-First Search (BFS) is guaranteed to find the shortest path in an unweighted graph, but its main drawback is its potentially high memory requirement.

Timer  30 Seconds



Question - 16

Fill the gap with a suitable answer:

A counting problem asks for the total number of possible solutions, rather than finding one specific solution itself.

Timer  30 Seconds



Question - 17

Fill the gap with a suitable answer:

Breadth-First Search (BFS) is a uninformed search algorithm because it uses no problem-specific knowledge to guide the search, exploring all nodes at one level before moving to the next.

Timer  30 Seconds



Question - 18

Fill the gap with a suitable answer:

The goal of an optimization problem is to find the best possible solution, which either minimize a cost function or maximize a utility function.

Timer  30 Seconds



Question - 19

Fill the gap with a suitable answer:

A decision problem is a computational problem that can only have two possible answers: Yes / True or No / False.

Timer



30 Seconds



Question - 20

Fill the gap with a suitable answer:

A* search is an informed search algorithm because it uses a combined evaluation function, $f(n) = g(n) + h(n)$, where $g(n)$ is the cost from the start node and $h(n)$ is the heuristic estimate of the cost to the goal.

Timer  30 Seconds



Question -21

True or False:

A Decision Problem is always considered computationally 'easier' than its corresponding Optimization Problem.

TRUE

Timer  30 Seconds



Question -22

True or False:

Depth-First Search (DFS) is generally preferred over Breadth-First Search (BFS) for finding the shortest path in an unweighted graph.

FALSE

Timer  30 Seconds



Question -23

True or False:

A Counting Problem corresponding to a decision problem asks, "How many solutions exist?" rather than "Does a solution exist?".

TRUE

Timer  30 Seconds



Question -24

True or False:

Any Optimization Problem can be rephrased as a series of Decision Problems (e.g., "Is there a solution better than value K?")

TRUE

Timer  30 Seconds



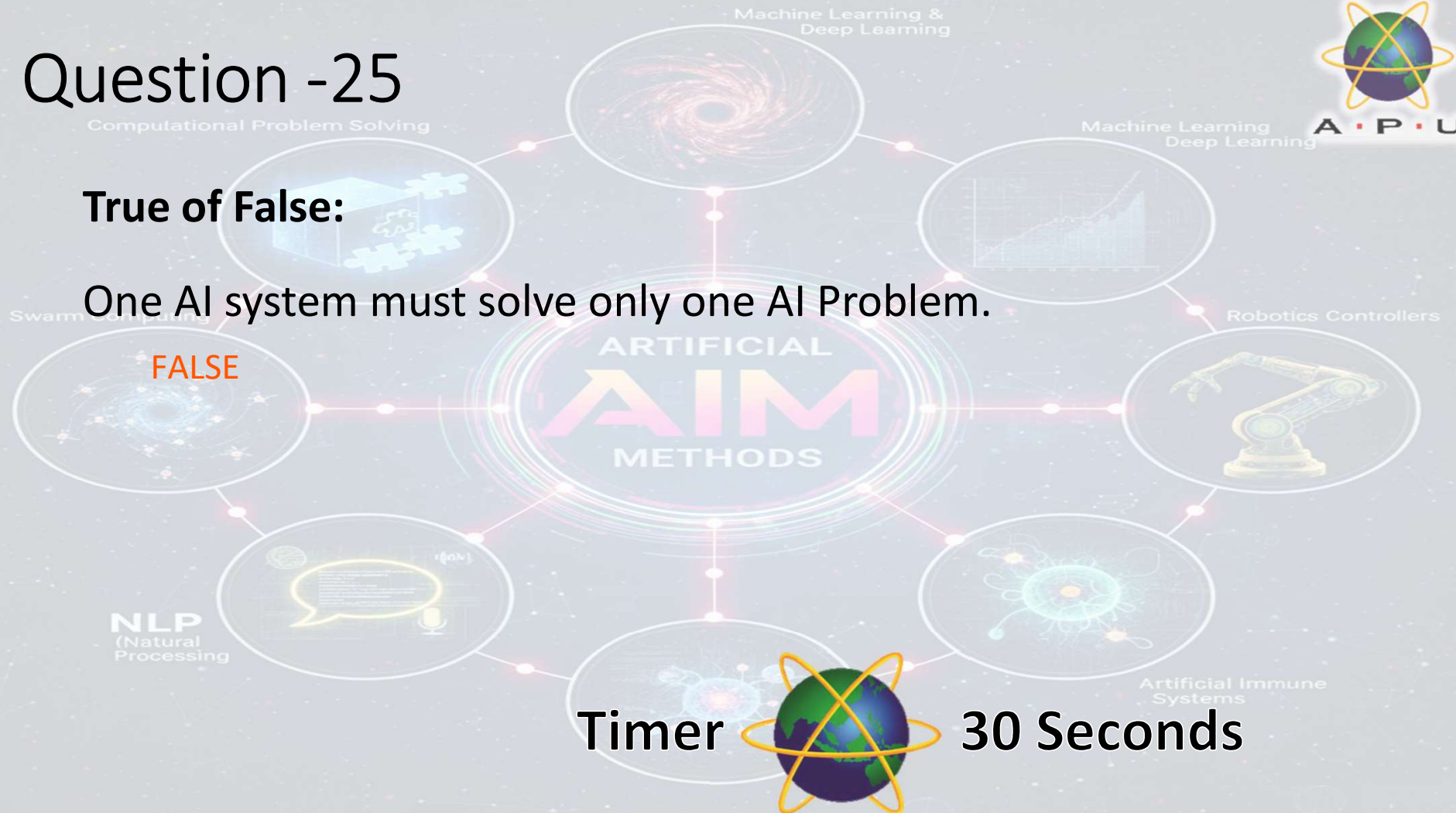
Question -25

True or False:

One AI system must solve only one AI Problem.

FALSE

Timer  30 Seconds



Next Class...

- We will learn about...
 - Learning Problems
 - Machine Learning





Q & A